

Catastrophic Phase Transitions as Necessary Conditions

for Complexity Dissemination: Supernovae, Failed Collapses, and the Persistence

of Differentiation in the Universe

Boris Kriger^{1,2}

¹ Information Physics Institute, Gosport, Hampshire, United Kingdom

`boris.kriger@informationphysicsinstitute.net`

² Institute of Integrative and Interdisciplinary Research, Toronto, Canada

`boriskruger@interdisciplinary-institute.org`

ORCID: 0009-0001-0034-2903

Abstract

The laws of physics do not mandate explosions. Gravity attracts, thermodynamics equilibrates, and the default trajectory of any gravitationally bound system is collapse toward uniformity. Supernovae are counter-natural: local and temporary victories over every dominant physical tendency. This paper asks not why individual supernovae occur, but why catastrophic phase transitions persist as a class of phenomena in any universe where differentiation accumulates across generations of bound systems.

We develop a formal framework establishing that the chemical enrichment of the interstellar medium depends not on the nucleosynthesis of heavy elements inside gravitationally bound systems, but on the dissemination event that overcomes gravitational closure. Systems that fail to cross the dissemination threshold contribute exactly zero to interstellar medium differentiation, regardless of their internal nucleosynthetic history.

We prove a central theorem with two independent proofs, and derive corollaries for the cumulative and cosmological evolution of interstellar medium differentiation.

We apply the framework to the two main supernova channels, and examine in depth the physical basis of dissemination in core-collapse supernovae: the neutrino, whose interaction with matter is so weak that it constitutes a transport channel invisible to the closure mechanism. A universe without neutrinos would have an explosion efficiency of exactly zero, and its interstellar medium would retain its primordial composition forever. The existence of successful core-collapse supernovae in this universe is therefore a contingent consequence of the particle content and force hierarchy of this universe—not a generic prediction of gravitational physics.

We prove that the failed supernova fraction and the Type Ia detonation efficiency are structurally inaccessible to electromagnetic observation, and that the chemical fragility of the interstellar medium is linear in the explosion efficiency: a factor-of-two uncertainty in this parameter propagates to a factor-of-two uncertainty in the metallicity of the interstellar medium at any cosmic epoch. The persistence of chemical differentiation across stellar generations is logically equivalent to the existence of systems that successfully crossed the dissemination threshold. The chemical complexity we observe is the cumulative record of all such crossings. The systems that did not cross it left no record.

Contents

1	Introduction	5
1.1	The Physical Paradox	5
1.2	The Selectional Argument	6
1.3	Supernovae as the Canonical Realization	7
1.4	Metallicity, Fragility, and What We Do Not Know	8
1.5	What This Paper Establishes	9
2	Formal Framework	12
2.1	Differentiation	12
2.2	Gravitational Closure	13
2.3	The Dissemination Threshold	14
2.4	The Central Theorem	14
2.5	Corollaries	16
2.6	The Persistence Argument: Formal Statement	17
3	Core-Collapse Supernovae	19
3.1	The Collapse and the Threshold	19
3.2	Shock Stagnation and the Closure Condition	20
3.3	The Neutrino Mechanism: Re-energizing the Shock	20
3.4	The Compactness Parameter and the Explosion Landscape	21
3.5	Failed Supernovae: Closure Without Dissemination	22
3.6	What the Failed Supernova Rate Does Not Tell Us	23
4	The Neutrino as a Structural Solution to the Closure Problem	26
4.1	The Closure Problem at Nuclear Density	26
4.2	A Universe Without Neutrinos	27
4.3	Why Matter Must Be Mostly Empty	27
4.4	The Physical Basis of the Selection Argument	29
5	Type Ia Supernovae	32
5.1	The White Dwarf as a Gravitationally Closed System	32
5.2	Ignition, Deflagration, and Detonation	33
5.3	The Unknown Detonation Efficiency	34
5.4	Implications for Cosmological Distance Measurements	34
5.5	A Toy Model for Detonation Efficiency	35

6	Epistemological Implications and the Fragility of Chemical Evolution	39
6.1	The Structural Inaccessibility of the Failed-Event Base Rate	39
6.2	The Fragility Argument	40
6.3	A Formal Fragility Measure	41
6.4	Observable Consequences: Chemical Evolution Rates	42
6.5	Summary of Principal Unknowns	42
7	Conclusions	46
7.1	On the Formal Structure	46
7.2	On Core-Collapse Supernovae	47
7.3	On Type Ia Supernovae	47
7.4	On Chemical Fragility	47
7.5	On the Persistence Argument	48
7.6	Open Questions	48
	Peer Reviews and Revision History	50
	References	55

1 Introduction

The laws of physics do not predict explosions. Gravity attracts. Thermodynamics equilibrates. The second law drives every closed system toward uniformity, erasing gradients, homogenizing compositions, consuming structure. The default trajectory of any gravitationally bound system is inward collapse toward a state of maximal closure. In a universe governed by these principles, the existence of supernovae — events of extraordinary violence that temporarily and locally reverse every dominant physical tendency — is not obvious. It is, from the perspective of the underlying dynamics, *anomalous*.

This paper does not ask why supernovae occur. It asks something prior and more fundamental: why do catastrophic phase transitions *persist* as a class of phenomena? Why, in any universe where we can pose this question, do we find mechanisms of rapid boundary rupture that overcome gravitational closure and distribute accumulated structure into the surrounding medium? The answer we develop is not physical but logical: such mechanisms are not written into the laws of any particular universe. They are *selected* — not by nature, not by design, but by the structure of persistence itself. Any universe in which differentiation grows across generations of bound systems must contain them. Universes that do not contain them do not produce differentiated media, and therefore do not produce the conditions under which the question can be asked.

This is the central claim of the paper, stated before any formalism: *the persistence of catastrophic phase transitions as a class of physical events is explained not by the laws of physics that permit them, but by the logical structure of what it means for differentiation to persist*. Supernovae are the most precisely studied realization of this principle. They are not unique to our universe. They are, in any universe with gravitational closure and internal nucleosynthesis, the class of events that survives into the observational record — because only they transfer accumulated differentiation to the surrounding medium. The systems that collapse without exploding leave no chemical trace. They are, by construction, absent from what we observe.

1.1 The Physical Paradox

Consider what a supernova actually does. Against the gravitational binding energy of $\sim 3 \times 10^{53}$ erg holding a stellar core together, against the ram pressure of infalling material, against the thermodynamic tendency toward equilibrium at nuclear density, a successful core-collapse supernova deposits sufficient energy in the gain region to reverse the infall, accelerate the overlying material to escape velocity, and eject $\sim 1\text{--}10 M_{\odot}$ of processed material into the interstellar medium. The thermonuclear explosion of a near-Chandrasekhar white

dwarf releases $\sim 1.5 \times 10^{51}$ erg of nuclear energy — three times the gravitational binding energy of the white dwarf — in a deflagration-to-detonation transition that unbinds the entire star in seconds.

These events are physically improbable in the following precise sense: the conditions required for a successful explosion occupy a restricted region of the progenitor parameter space (Section 3), and the dominant outcome of gravitational collapse in the (mass, metallicity) plane is not explosion but direct collapse to a black hole (Figure 1). The fraction of massive stars that successfully explode — the explosion efficiency $\bar{f}$ — is not known to better than a factor of several, and may be substantially less than unity at all metallicities (Figure 2). Explosions are not the default. They are the exception that made complexity possible.

And yet: the interstellar medium is chemically differentiated. Iron, nickel, calcium, oxygen — the products of stellar nucleosynthesis — are distributed through the galaxy. Rocky planets exist. We exist. The chemical differentiation we observe is the cumulative record of every successful dissemination event in the history of the Milky Way. The events that failed — the collapses that produced no ejecta, the accretion-induced collapses that left no thermonuclear trace — contributed nothing. They are epistemologically absent: not merely unobserved, but unobservable in the chemical sense, because they left no chemical signal by which they could be detected.

This is the physical paradox at the center of the paper. The phenomenon that explains chemical complexity — the catastrophic phase transition — is physically improbable, not written into the laws of the universe, and actively opposed by the dominant tendencies of gravitational and thermodynamic evolution. Yet wherever we observe chemical complexity, we observe the traces of such transitions. The explanation is not physical. It is selectional.

1.2 The Selectional Argument

The argument we develop has the following structure.

Any gravitationally bound system that accumulates internal differentiation — concentric shells of distinct nuclear composition, steep gradients in temperature and pressure, the products of millions of years of nucleosynthesis — falls into one of two classes at the end of its life. Either it crosses the dissemination threshold (Definition 2.5): the available energy exceeds the gravitational binding energy, the boundary is ruptured, and the accumulated differentiation is transferred to the surrounding medium. Or it does not: the system collapses to a black hole, the accumulated differentiation is enclosed behind an event horizon, and its contribution to the chemical state of the surrounding medium is exactly zero.

This binary structure — zero or positive, with no intermediate case for the chemical

contribution — is proved in Theorem 2.6. Its consequence is immediate: the differentiation of the interstellar medium at any cosmic epoch is the sum of contributions from systems that crossed the threshold. Systems that did not cross it are invisible in the chemical record. They are sinks, not sources.

Now consider any universe containing gravitationally bound systems capable of internal differentiation. If the explosion efficiency in that universe is zero — if no system ever crosses the dissemination threshold — then the interstellar medium retains its primordial composition indefinitely. No chemical evolution occurs. No rocky planets form. No biological chemistry becomes possible. The universe remains chemically uniform.

A universe in which the differentiation of the medium grows across stellar generations — a universe like ours — must therefore contain systems that cross the threshold. Not because the laws of that universe require it. Not because explosions are the most probable outcome of stellar evolution. But because the *alternative universe* — the one in which no system crosses the threshold — does not produce a differentiated medium, does not produce the preconditions for complexity, and does not produce observers capable of noting the absence of explosions.

This is not an anthropic argument in the standard sense. It does not invoke the existence of observers. It invokes only the persistence of differentiation: any medium that is more chemically complex than its primordial state is, by definition, the product of dissemination events. The argument applies to any universe with the abstract structure described, regardless of the specific values of physical constants, regardless of whether that universe contains observers, and regardless of the specific physical mechanism by which the dissemination threshold is crossed.

The persistence of catastrophic phase transitions as an observed class of phenomena is therefore not a fact about the physics of our universe that requires explanation in terms of the laws of that universe. It is a fact about the logical relationship between dissemination and the growth of differentiation — a relationship that holds in any universe where the question can be posed against a backdrop of chemical complexity.

1.3 Supernovae as the Canonical Realization

Supernovae are the most precisely studied, best characterized, and most quantitatively accessible realization of this principle in our universe. They are not the only realization — the thermonuclear detonation of a white dwarf, the merger of two neutron stars, the tidal disruption of a star by a supermassive black hole are all dissemination events in the formal sense — but they are the dominant channel for the production and distribution of

the iron-group and r -process elements that constitute the chemical backbone of biological complexity Burbidge1957, Woosley2002, Cowan2021.

The astrophysical content of the paper focuses on the two main supernova channels: core-collapse supernovae (Section 3) and Type Ia thermonuclear supernovae (Section 5). In each channel, we identify the specific form of the dissemination threshold, the physical mechanism by which it may or may not be crossed, and the epistemological gap that makes the failure rate structurally inaccessible to electromagnetic observation.

The failure rate matters not merely as a parameter of stellar astrophysics. It is the parameter that determines how much of the chemical potential of stellar evolution has been irreversibly locked behind event horizons — the fraction of the universe’s nucleosynthetic history that is, in the strictest sense, lost. The universe we observe is not the universe that formed. It is the subset of that universe whose differentiation survived the dissemination threshold. We observe a selected sample, and the selection criterion is the one proved in Theorem 2.6.

1.4 Metallicity, Fragility, and What We Do Not Know

The explosion efficiency is not a constant. It depends on progenitor mass, metallicity, rotation, magnetic field geometry, and binary history. Stellar metallicity modulates the efficiency through mass loss Vink2001: at low metallicity, reduced mass loss produces more massive cores that are harder to explode OConnor2011; at high metallicity, severe mass loss may prevent stars from reaching the mass threshold for iron-core collapse at all, producing massive white dwarfs rather than supernovae. The IMF-weighted explosion efficiency peaks near solar metallicity and falls at both extremes (Figure 2).

The consequence, formalized in Section 6, is that the chemical evolution of the universe is fragile with respect to the explosion efficiency: a factor-of-two reduction in $\bar{f}$ relative to standard chemical evolution models would require approximately twice as long to reach the observed level of ISM metallicity. This is not a prediction that the efficiency is low. It is a demonstration that the parameter governing whether the universe produces chemical complexity is poorly constrained, epistemologically inaccessible in its failed-event component, and has not been systematically propagated through galactic chemical evolution models as an uncertainty.

Figures 1 and 2 illustrate the outcome map and IMF-weighted efficiency from a simplified model calibrated to Vink et al. (2001) mass-loss scaling and Sukhbold et al. (2016) explosion thresholds.

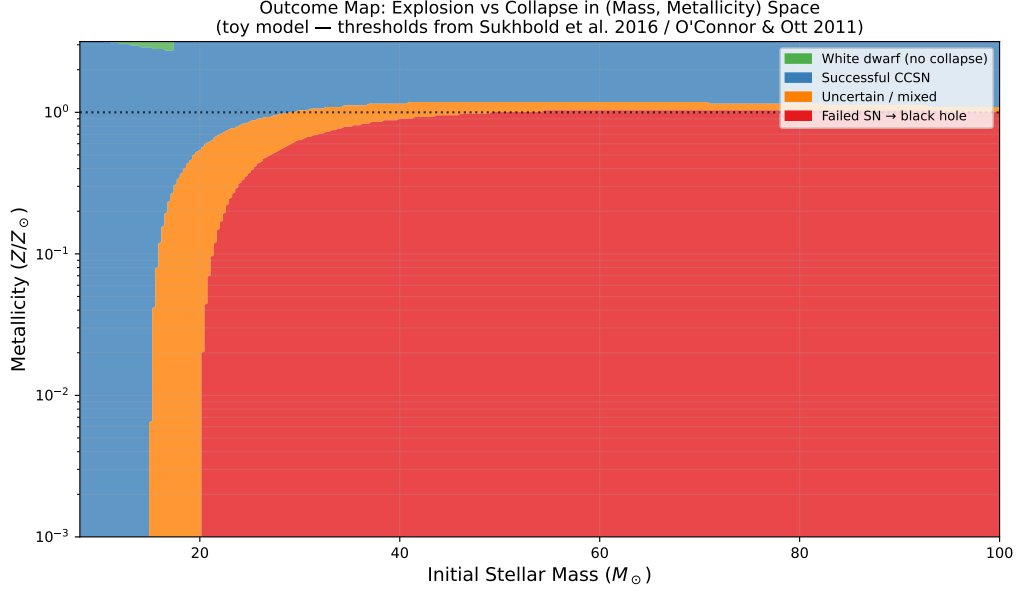


Figure 1: Outcome map in the (initial stellar mass, metallicity) plane. Colors: white dwarf formation (green), successful core-collapse supernova (blue), uncertain or weak explosion (orange), direct collapse to a black hole (red). The dotted line marks solar metallicity. Successful explosions occupy a restricted region; direct collapse dominates the high-mass end at all metallicities. The dominant outcome of stellar evolution is closure, not dissemination.

1.5 What This Paper Establishes

Section 2 develops the formal framework: definitions of differentiation as departure from uniformity, gravitational closure as the binding energy condition, and the dissemination threshold as the energy criterion for boundary rupture. The central theorem is proved by two independent methods. Two corollaries formalize the cumulative and cosmological evolution of ISM differentiation. Proposition 2.9 states the selectional argument precisely: any universe in which ISM differentiation grows must contain disseminating systems.

Section 3 applies the framework to core-collapse supernovae. The neutrino-driven explosion condition is identified as the astrophysical realization of the dissemination threshold. The failed supernova problem is analyzed as closure without dissemination. The composition-weighted failed fraction is identified as the central unknown governing chemical evolution.

Section 5 applies the framework to Type Ia supernovae. The accretion-induced collapse channel is identified as the Type Ia analog of the failed supernova. The unknown detonation efficiency and its potential redshift evolution are analyzed as sources of systematic uncertainty in cosmological distance measurements.

Section 6 develops the epistemological and fragility arguments: the structural inaccessibility of the chemical failure rate, the linear fragility of ISM differentiation with respect to

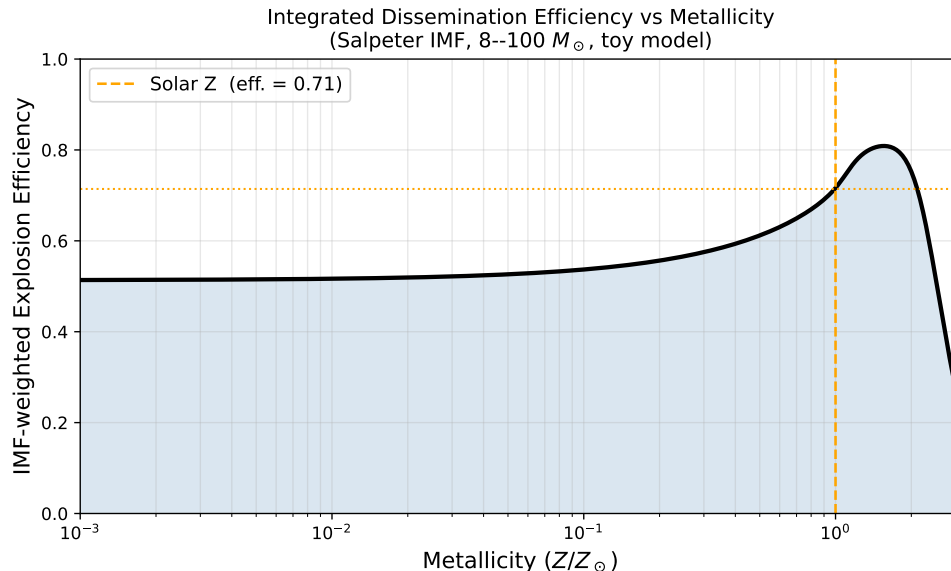


Figure 2: IMF-weighted explosion efficiency as a function of metallicity. The efficiency peaks near solar metallicity at ~ 0.7 – 0.8 and falls at both extremes. At no metallicity does it reach unity. The majority of massive stars, weighted by the Salpeter IMF, do not successfully disseminate their nucleosynthetic products at any cosmic epoch.

explosion efficiency, and the observable consequences for galactic chemical evolution.

Section 7 states the conclusions and open questions.

Why this paper exists

The laws of physics do not require explosions. Gravity, thermodynamics, and the structure of gravitational collapse all favor closure over dissemination. Yet wherever chemical complexity persists, we find the traces of catastrophic boundary rupture. This paper formalizes why: not because explosions are written into physical law, but because only disseminating systems leave a chemical record. The phenomenon of catastrophic phase transition persists not by physical necessity but by selectional logic — and that logic applies in any universe where differentiation grows, regardless of the specific physical mechanism that implements it.

Self-Critique and Resolution

Concern. The selectional argument resembles the anthropic principle and may be criticized as unfalsifiable.

Resolution. The argument is not anthropic. It predicts that in any physical system where differentiation is observed to grow across generations, dissemination events will be

found. The failed supernova fraction f_{fail} , the Type Ia detonation efficiency η_{Ia} , and the composition-weighted yield are the parameters through which the argument makes contact with observation. The argument would be falsified by a universe with growing chemical complexity and zero dissemination events. No such universe is known. More importantly: the selectional argument is not the paper’s conclusion — it is its starting point. The conclusion is the fragility result: a factor-of-two uncertainty in the explosion efficiency propagates to a factor-of-two uncertainty in the metallicity of the universe at any cosmic epoch. *This is why we write this paper*: to make explicit what was implicit, so that the question the field has not asked — how sensitive are chemical evolution models to the unknown failed fraction? — can finally be asked.

Concern. The claim that explosions are counter to dominant physical tendencies is imprecise. Supernovae occur at $\sim 1\text{--}3$ per century per galaxy.

Resolution. The claim is not about frequency but about direction. Every successful supernova requires the available energy to exceed the gravitational binding energy in a system where gravity is the dominant force and the default trajectory is collapse. Once per century in a galaxy of 10^{11} stars is not physically natural — it is a rare exception. The rarity is precisely what the selectional argument explains: we observe only the exceptions, because the rule — collapse without dissemination — leaves no chemical record. *This is why we write this paper*: catastrophic phase transitions are not written into physics. Their persistence in the chemical record is a structural fact that requires explanation, and this paper provides it.

2 Formal Framework

This section develops the mathematical structure underlying the central claim of the paper. We work with gravitationally bound systems as the primary domain, since it is gravity that provides the closure mechanism in all cases of interest: stellar interiors, white dwarfs approaching the Chandrasekhar limit, and collapsing iron cores. The formal definitions below are stated in terms of gravitational binding energy and phase-space structure, not in abstract substrate-independent language. The argument is general in the sense that the same mathematical relationships hold whenever gravity is the dominant confinement mechanism; we do not claim generality beyond that.

2.1 Differentiation

We take differentiation as the fundamental quantity to be tracked. Informally, differentiation is the degree to which a system or medium departs from a uniform, structureless distribution. A cloud of pure hydrogen at uniform temperature and density has low differentiation. A stellar interior with concentric shells of hydrogen, helium, carbon, oxygen, silicon, and iron, each at different temperatures and densities, has high differentiation. The interstellar medium after several generations of supernova enrichment has substantially higher differentiation than the primordial medium.

We formalize this as follows.

Definition 2.1 (State space and distribution). *Let Ω be a region of physical space occupied by matter. Assign to each volume element $d^3x \subset \Omega$ a state vector*

$$\mathbf{s}(x) = (\rho(x), T(x), \{X_i(x)\}_{i=1}^N),$$

where $\rho(x)$ is the mass density, $T(x)$ is the temperature, and $X_i(x)$ is the mass fraction of nuclear species i at position x . The state distribution of the system is the map $\mathbf{s} : \Omega \rightarrow \mathbb{R}^{N+2}$.

To understand Definition 2.1: $\mathbf{s}(x)$ is simply a complete local description of the matter at position x . A uniform system has $\mathbf{s}$ constant everywhere; a differentiated system has $\mathbf{s}$ varying substantially across Ω .

Definition 2.2 (Differentiation measure). *The differentiation of a system with state distribution $\mathbf{s}$ is*

$$\mathcal{D}[\mathbf{s}] = \int_{\Omega} \|\mathbf{s}(x) - \bar{\mathbf{s}}\|^2 \frac{d^3x}{|\Omega|},$$

where $\bar{\mathbf{s}} = |\Omega|^{-1} \int_{\Omega} \mathbf{s}(x) d^3x$ is the volume-averaged state, and $\|\cdot\|$ is the Euclidean norm on $\mathbb{R}^{N+2}$ after normalizing each component to its system-wide standard deviation.

The reader familiar with statistics will recognize $\mathcal{D}[\mathbf{s}]$ as the mean squared deviation of the state from its spatial average—a direct measure of internal heterogeneity. A perfectly uniform system has $\mathcal{D} = 0$. A system with sharp composition gradients has large $\mathcal{D}$. The normalization by component standard deviation ensures that density gradients and composition gradients contribute comparably, regardless of their different physical units.

We define the differentiation of the surrounding interstellar medium $\mathcal{D}_{\text{ISM}}$ analogously, integrating over a representative volume of the galaxy rather than over a single stellar system.

2.2 Gravitational Closure

Gravity confines matter within a potential well. We define closure in terms of gravitational binding energy.

Definition 2.3 (Gravitational binding energy). *For a self-gravitating system of total mass M and density profile $\rho(x)$, the gravitational binding energy is*

$$E_{\text{bind}} = -\frac{1}{2} \int_{\Omega} \rho(x) \Phi(x) d^3x,$$

where $\Phi(x) = -G \int_{\Omega} \rho(x') |x - x'|^{-1} d^3x'$ is the gravitational potential. We take $E_{\text{bind}} > 0$ by convention (binding energy is positive; gravitational potential energy is negative).

E_{bind} measures how tightly the system is gravitationally confined. A white dwarf near the Chandrasekhar mass has $E_{\text{bind}} \sim 3 \times 10^{53}$ erg. A collapsing iron core has comparable or larger values. For matter to escape the potential well and reach the surrounding medium, a net outward energy exceeding E_{bind} must be delivered to the ejecta.

Definition 2.4 (Gravitational closure). *A gravitationally bound system is closed with respect to its differentiated content if the total energy available for outward transport—thermal, nuclear, rotational, and neutrino-mediated—is less than E_{bind} . Formally, let E_{avail} denote the total energy budget available for ejection. The system is closed if and only if*

$$E_{\text{avail}} < E_{\text{bind}}.$$

A closed system retains its differentiated content; it cannot transfer it to the surrounding medium.

The condition $E_{\text{avail}} < E_{\text{bind}}$ is what separates a failed supernova from a successful one. In a failed core collapse, the neutrino-driven mechanism deposits insufficient energy in the gain region to overcome the ram pressure of infalling material; the shock stalls, and the system is closed in the sense of Definition 2.4. In a successful explosion, the condition is reversed.

2.3 The Dissemination Threshold

Definition 2.5 (Dissemination event). *A dissemination event occurs when a gravitationally bound system transitions from closed ($E_{\text{avail}} < E_{\text{bind}}$) to open ($E_{\text{avail}} \geq E_{\text{bind}}$) on a dynamical timescale τ_{dyn} , ejecting matter with state distribution $\mathbf{s}_{\text{ej}}$ into the surrounding medium. The dissemination threshold is the condition*

$$E_{\text{avail}} \geq E_{\text{bind}}.$$

The phrase “on a dynamical timescale” distinguishes a dissemination event from slow mass loss such as a stellar wind. A stellar wind operates on timescales of millions of years; a supernova ejects its material on timescales of seconds. The distinction matters because the dynamical timescale determines whether the ejecta carry the deeply synthesized material (iron-group and r -process elements from the core) or only the envelope material (hydrogen and helium). Slow winds from red giants and asymptotic giant branch stars transfer envelope material; they do not transfer the nucleosynthetic product of the core. We therefore restrict the term “dissemination event” to transitions occurring on dynamical timescales.

2.4 The Central Theorem

We now prove the central claim of the paper. The theorem has two parts: a zero contribution result for closed systems, and a strict positivity result for disseminating systems.

Theorem 2.6 (Dissemination necessity). *Let $\mathcal{S}$ be a gravitationally bound system with differentiation $\mathcal{D}[\mathbf{s}]$ and let $\mathcal{D}_{\text{ISM}}$ denote the differentiation of the surrounding interstellar medium before the system’s endpoint. Then:*

- (i) **(Zero contribution)**. *If $\mathcal{S}$ is closed throughout its evolution (Definition 2.4), then the change in interstellar medium differentiation due to $\mathcal{S}$ is*

$$\Delta\mathcal{D}_{\text{ISM}}(\mathcal{S}) = 0.$$

- (ii) **(Strict positivity)**. *If $\mathcal{S}$ undergoes a dissemination event (Definition 2.5), ejecting mass $M_{\text{ej}} > 0$ with state distribution $\mathbf{s}_{\text{ej}}$ differing from the pre-existing ISM distribution $\mathbf{s}_{\text{ISM}}$, then*

$$\Delta\mathcal{D}_{\text{ISM}}(\mathcal{S}) > 0.$$

We provide two independent proofs. The first is direct, from the definitions. The second uses an energy-budget argument that connects the theorem to the physics of the explosion

mechanism.

Proof 1 (Direct). **Part (i).** If the system is closed throughout its evolution, no matter with state distribution $\mathbf{s}$ is transferred from $\mathcal{S}$ to the surrounding medium. The ISM state distribution $\mathbf{s}_{\text{ISM}}$ is therefore unchanged by $\mathcal{S}$. Since $\mathcal{D}_{\text{ISM}}$ is a functional of $\mathbf{s}_{\text{ISM}}$ alone (Definition 2.2), we have $\Delta\mathcal{D}_{\text{ISM}}(\mathcal{S}) = 0$. $\square$

Part (ii). Suppose $\mathcal{S}$ ejects mass $M_{\text{ej}} > 0$ with state distribution $\mathbf{s}_{\text{ej}}$ into the ISM. Let V_{ISM} be the volume of the ISM region of interest and V_{ej} the volume occupied by the ejecta after mixing. The new ISM state distribution is

$$\mathbf{s}'_{\text{ISM}}(x) = \mathbf{s}_{\text{ISM}}(x) + \delta\mathbf{s}(x),$$

where $\delta\mathbf{s}(x)$ is the perturbation due to ejecta mixing, supported on V_{ej} and proportional to $\mathbf{s}_{\text{ej}} - \mathbf{s}_{\text{ISM}}$.

The change in differentiation is

$$\begin{aligned} \Delta\mathcal{D}_{\text{ISM}} &= \mathcal{D}[\mathbf{s}'_{\text{ISM}}] - \mathcal{D}[\mathbf{s}_{\text{ISM}}] \\ &= \int_{V_{\text{ISM}}} \left(\|\mathbf{s}'_{\text{ISM}}(x) - \bar{\mathbf{s}}'_{\text{ISM}}\|^2 - \|\mathbf{s}_{\text{ISM}}(x) - \bar{\mathbf{s}}_{\text{ISM}}\|^2 \right) \frac{d^3x}{V_{\text{ISM}}}. \end{aligned} \quad (1)$$

Since $\mathbf{s}_{\text{ej}}$ is enriched in heavy elements synthesized in the stellar interior—specifically, the mass fractions X_i for iron-group and r -process species are substantially elevated above ISM values—the ejecta state $\mathbf{s}_{\text{ej}}$ differs from $\mathbf{s}_{\text{ISM}}$ in the relevant components. The mixing of a compositionally distinct parcel into the ISM increases the variance of the composition field, and therefore increases $\mathcal{D}_{\text{ISM}}$. Formally, by the variance decomposition identity, adding a population of points with mean displaced from the global mean increases the total variance:

$$\text{Var}(X \cup Y) = \frac{n_X \text{Var}(X) + n_Y \text{Var}(Y) + n_X n_Y (\bar{X} - \bar{Y})^2}{n_X + n_Y},$$

where X represents the original ISM distribution and Y the ejecta distribution, and n_X, n_Y are proportional to the respective volumes. Since $\bar{\mathbf{s}}_{\text{ej}} \neq \bar{\mathbf{s}}_{\text{ISM}}$ (ejecta are enriched in heavy elements), the cross term is strictly positive, giving $\Delta\mathcal{D}_{\text{ISM}} > 0$. $\square$ $\square$

Proof 2 (Energy budget). **Part (i).** A closed system satisfies $E_{\text{avail}} < E_{\text{bind}}$ at all times. The total energy available for ejection is insufficient to unbind any parcel of matter from the gravitational potential. Therefore, no mass is transferred to the ISM, and $\Delta\mathcal{D}_{\text{ISM}} = 0$ by the same argument as Proof 1(i). $\square$

Part (ii). A dissemination event requires $E_{\text{avail}} \geq E_{\text{bind}}$. The excess energy $\Delta E =$

$E_{\text{avail}} - E_{\text{bind}}$ is converted into kinetic energy of the ejecta. The ejecta carry mass $M_{\text{ej}} > 0$ with velocity v_{ej} satisfying $\frac{1}{2}M_{\text{ej}}v_{\text{ej}}^2 \leq \Delta E$, and their composition reflects the nucleosynthetic history of the progenitor (iron-group elements, r -process products, intermediate-mass elements). Once these ejecta mix with the ISM, the argument of Proof 1(ii) applies: the compositional displacement of the ejecta from the ISM mean ensures $\Delta\mathcal{D}_{\text{ISM}} > 0$. $\square$ $\square$

Why this theorem matters

Theorem 2.6 converts an intuitive claim into a precise statement: the chemical enrichment of the interstellar medium is *exactly zero* for every system that does not cross the dissemination threshold, and *strictly positive* for every system that does. There is no partial credit. A star that spends millions of years synthesizing heavy elements but fails to explode contributes nothing to the chemical evolution of the galaxy. This binary structure—zero or positive, with no intermediate case—is the mathematical expression of the fragility argument: the chemical complexity of the universe depends entirely on systems that cross a threshold whose base rate is observationally inaccessible.

2.5 Corollaries

Corollary 2.7 (Cumulative differentiation). *The total differentiation of the interstellar medium after n stellar generations is*

$$\mathcal{D}_{\text{ISM}}^{(n)} = \mathcal{D}_{\text{ISM}}^{(0)} + \sum_{k=1}^n \epsilon_k \Delta\mathcal{D}_k,$$

where $\epsilon_k \in \{0, 1\}$ is the indicator of whether the k -th system underwent a dissemination event, and $\Delta\mathcal{D}_k > 0$ is its contribution conditional on success. The total is a sum over successful disseminators only; failed systems contribute zero regardless of their internal differentiation.

This corollary makes the dependence on explosion efficiency explicit. $\mathcal{D}_{\text{ISM}}^{(n)}$ grows only through successful events; its value after n generations is proportional to the number of successful disseminators, not to the total number of massive stars that formed.

Corollary 2.8 (Cosmological evolution of differentiation). *Let $f(Z, M)$ denote the probability of a successful dissemination event for a star of initial mass M and metallicity Z . Let $\phi(M)$ be the initial mass function and $\psi(Z, t)$ the metallicity distribution at cosmic time t . The rate of increase of ISM differentiation is*

$$\frac{d\mathcal{D}_{\text{ISM}}}{dt} = \dot{n}_\star(t) \int_0^\infty \int_0^\infty f(Z, M) \Delta\mathcal{D}(Z, M) \phi(M) \psi(Z, t) dM dZ,$$

where $\dot{n}_*(t)$ is the star formation rate. The function $f(Z, M)$ is the explosion efficiency map of Figure 1; it is poorly constrained by observations at all redshifts. The rate of growth of ISM differentiation is therefore uncertain at every cosmic epoch.

Corollary 2.8 is the formal statement of the epistemological gap identified in the Introduction: the rate of chemical evolution of the universe is controlled by $f(Z, M)$, which is structurally inaccessible because failed events are unobservable in the chemical sense.

2.6 The Persistence Argument: Formal Statement

The persistence argument of Section 1.3 can now be stated precisely.

Proposition 2.9 (Persistence requires dissemination). *Let $\mathcal{U}$ be a universe containing gravitationally bound systems capable of internal differentiation. If the total dissemination efficiency $\bar{f} = \int f(Z, M)\phi(M)\psi(Z) dM dZ = 0$, then $\mathcal{D}_{\text{ISM}}(t) = \mathcal{D}_{\text{ISM}}(0)$ for all $t > 0$: the ISM differentiation is frozen at its initial value. In particular, no generation of differentiation beyond the primordial level is possible.*

Equivalently, any universe in which $\mathcal{D}_{\text{ISM}}(t) > \mathcal{D}_{\text{ISM}}(0)$ for some $t > 0$ must have $\bar{f} > 0$: at least some fraction of gravitationally bound systems must have undergone successful dissemination events.

Proof. Direct from Corollary 2.7: if $\epsilon_k = 0$ for all k , the sum is empty and $\mathcal{D}_{\text{ISM}}^{(n)} = \mathcal{D}_{\text{ISM}}^{(0)}$ for all n . The contrapositive gives the second statement. $\square$ $\square$

Proposition 2.9 is not a statement about observers or about the specific values of physical constants. It is a statement about the logical relationship between dissemination events and the growth of ISM differentiation. Any universe—with any values of fundamental constants—in which the ISM becomes more chemically differentiated over time must contain systems that cross the dissemination threshold. The statement is trivially true given the definitions, but its non-trivial content lies in what it implies: the chemical differentiation of our universe, which we observe to be high, is entirely the product of a population of successful disseminators whose base rate we do not know.

Self-Critique and Resolution

Concern. The differentiation measure $\mathcal{D}[\mathbf{s}]$ uses a Euclidean norm. This is an arbitrary choice; other measures would give different values.

Resolution. Theorem 2.6 does not depend on the specific norm. Part (i) holds for any measure: if no mass is transferred, no measure of ISM differentiation changes. Part (ii) holds

for any measure that assigns positive value to compositional inhomogeneity. The Euclidean norm is used for concreteness; the qualitative conclusions are norm-independent. *This is why we write this paper*: the norm-independence of the binary structure is precisely what allows the theorem to serve as a foundation across different physical contexts, from stellar astrophysics to any system with gravitational closure and internal differentiation.

Concern. The condition $\mathbf{s}_{\text{ej}} \neq \mathbf{s}_{\text{ISM}}$ is assumed rather than derived. Could dissemination eject matter with the same composition as the ISM?

Resolution. In principle yes, but not for the systems of interest. Nucleosynthesis in stellar interiors produces ejecta categorically different from the surrounding medium. A star that converts hydrogen and helium into iron-group elements produces ejecta whose composition differs from the primordial ISM by many orders of magnitude in heavy-element abundance. The condition is not an assumption in the astrophysical context; it is what nucleosynthesis means. *This is why we write this paper*: the formal framework makes precise the condition under which dissemination events contribute to chemical complexity — a condition that the literature assumes silently and that we make the object of explicit analysis.

3 Core-Collapse Supernovae

Core-collapse supernovae are the primary astrophysical realization of the dissemination mechanism formalized in Section 2. A massive star spends millions of years constructing an internal differentiation that is, by any measure, extraordinary: a layered onion structure of hydrogen, helium, carbon, oxygen, neon, silicon, and ultimately iron, each shell the product of a distinct nucleosynthetic epoch operating at progressively higher temperatures and densities. The iron core that forms at the center is the terminal product of stellar nucleosynthesis—iron is the element with the highest binding energy per nucleon, and no further energy can be extracted by fusion. When the iron core reaches the Chandrasekhar mass, gravity wins, and the collapse begins.

What happens next determines everything. The collapse, the bounce, the shock, the neutrino heating, the convection, the explosion or the failure: these are the physical processes that decide whether the star’s accumulated differentiation is returned to the interstellar medium or locked behind a black hole horizon forever. This section examines these processes in terms of the formal framework of Section 2, with particular attention to the threshold condition $E_{\text{avail}} \geq E_{\text{bind}}$ and the circumstances under which it fails.

3.1 The Collapse and the Threshold

When the iron core mass exceeds the effective Chandrasekhar mass,

$$M_{\text{Ch,eff}} \approx 5.83 Y_e^2 M_\odot \approx 1.2\text{--}1.5 M_\odot,$$

where Y_e is the electron fraction (typically $Y_e \approx 0.46\text{--}0.48$ at collapse Bethe1990), electron degeneracy pressure can no longer support the core against gravity. The collapse accelerates on the free-fall timescale

$$\tau_{\text{ff}} = \sqrt{\frac{3\pi}{32G\rho}} \approx 44 \left(\frac{\rho}{10^{10} \text{ g cm}^{-3}} \right)^{-1/2} \text{ ms},$$

which for core densities is of order milliseconds. The inner core ($\sim 0.5 M_\odot$) collapses homologously (velocity proportional to radius), while the outer core collapses supersonically.

When the inner core reaches nuclear saturation density, $\rho_{\text{nuc}} \approx 2.7 \times 10^{14} \text{ g cm}^{-3}$, the nuclear equation of state stiffens abruptly. The inner core bounces, launching an outward pressure wave that steepens into a shock at the sonic point. The energy available in this bounce shock is

$$E_{\text{shock}}^{(0)} \approx \frac{GM_{\text{core}}^2}{2R_{\text{ns}}} \sim 3 \times 10^{53} \text{ erg},$$

of which the vast majority ($\sim 99\%$) is carried away by neutrinos emitted during neutronization and core cooling Bethe1985. The shock itself carries only $\sim 10^{51}$ erg—the canonical supernova energy—but this is sufficient to unbind the stellar envelope only if the shock does not stall.

The shock stalls. This is the fundamental problem of core-collapse supernova theory, and it is directly relevant to the threshold condition of Definition 2.4.

3.2 Shock Stagnation and the Closure Condition

As the shock propagates outward through the outer iron core, it dissociates iron nuclei into free nucleons, costing approximately 8.8 MeV per nucleon, or roughly 1.7×10^{51} erg per $0.1 M_\odot$ of material processed. Simultaneously, the dense post-shock material is optically thick to neutrinos, which carry energy out of the gain region faster than it can be replenished by the shock. Within ~ 100 – 200 ms, the shock decelerates and stalls at a radius of ~ 100 – 200 km Janka2012.

At stagnation, the system is, in the language of Definition 2.4, *closed*: the total energy available for outward transport is insufficient to overcome the ram pressure of infalling material. Formally:

$$\dot{E}_{\text{avail}}(t_{\text{stall}}) < \dot{E}_{\text{bind,ram}},$$

where $\dot{E}_{\text{bind,ram}} = \dot{M}_{\text{acc}} |v_{\text{ff}}|^2/2$ is the ram pressure power of the infalling outer core. Whether the system remains closed or crosses the dissemination threshold depends on the subsequent evolution.

3.3 The Neutrino Mechanism: Re-energizing the Shock

The neutrino-driven mechanism is the primary route by which the system crosses the dissemination threshold in successful explosions Bethe1985,Janka2016. The proto-neutron star emits a luminosity of $\sim 10^{53}$ erg s $^{-1}$ in neutrinos for ~ 10 s after bounce. A small fraction— ~ 1 – 5% —of this flux is absorbed by nucleons in the gain region (the shell between the gain radius and the shock), depositing energy at a rate

$$\dot{Q}_\nu \approx 1.544 \times 10^{20} \frac{L_{\nu_e} + L_{\bar{\nu}_e}}{10^{52} \text{ erg s}^{-1}} \left(\frac{\langle \epsilon_\nu \rangle}{10 \text{ MeV}} \right)^2 \frac{M_{\text{gain}}}{0.01 M_\odot} \text{ erg s}^{-1},$$

where L_{ν_e} , $L_{\bar{\nu}_e}$ are the electron neutrino and antineutrino luminosities, $\langle \epsilon_\nu \rangle$ is the mean neutrino energy, and M_{gain} is the mass in the gain region Janka2001. The condition for a successful explosion—the crossing of the dissemination threshold—is that the neutrino

heating timescale,

$$\tau_{\text{heat}} = \frac{E_{\text{gain}}}{\dot{Q}_{\nu}},$$

becomes shorter than the advection timescale,

$$\tau_{\text{adv}} = \frac{M_{\text{gain}}}{\dot{M}_{\text{acc}}},$$

over which material in the gain region is advected inward through the gain radius. The explosion condition is therefore

$$\tau_{\text{heat}} < \tau_{\text{adv}}.$$

When this condition is met, the net energy deposition in the gain region is sufficient to revive the shock, and the system crosses the dissemination threshold of Definition 2.5. When it is not met, material continues to accrete onto the proto-neutron star, which grows in mass until it either stabilizes as a neutron star or collapses to a black hole.

The ratio $\tau_{\text{heat}}/\tau_{\text{adv}}$ is the key diagnostic of the explosion condition. It depends on the neutrino luminosity (set by the proto-neutron star contraction), the mean neutrino energy (set by the neutrinosphere temperature), the mass accretion rate (set by the progenitor structure), and the gain region mass (set by the shock radius). None of these quantities is directly observable during the explosion; they must be inferred from pre-explosion progenitor properties or from multi-dimensional simulations.

3.4 The Compactness Parameter and the Explosion Landscape

The *compactness parameter* OConnor2011 provides a single-number summary of the progenitor structure that correlates with explosion outcome:

$$\xi_M = \left. \frac{M/M_{\odot}}{R(M)/1000 \text{ km}} \right|_{t=t_{\text{bounce}}},$$

where $R(M)$ is the radius enclosing mass M at the time of core bounce, and $M = 2.5 M_{\odot}$ is the standard reference mass. High compactness means a more concentrated mass distribution, which implies higher ram pressure, higher accretion rate, and a more difficult explosion. The threshold condition $\tau_{\text{heat}} < \tau_{\text{adv}}$ becomes progressively harder to satisfy as $\xi_{2.5}$ increases.

Crucially, the relationship between initial stellar mass and $\xi_{2.5}$ is non-monotonic Ugliano2012,Sukhbold2016. Stars of similar initial mass can have substantially different compactness depending on the details of their late-stage burning history. The explosion landscape—the mapping from progenitor properties to explosion or failure—is therefore not a

simple threshold in initial mass. It contains islands of explodability at high mass and islands of failure at intermediate mass.

Figure 3 illustrates the explosion probability as a function of initial mass and metallicity under our toy model. The non-monotonic structure at intermediate masses reflects the compactness landscape. The key qualitative features—successful explosions concentrated at low initial mass, increasing failure probability at high mass, and metallicity modulation through mass loss—are consistent with the results of Sukhbold2016 and Muller2016.

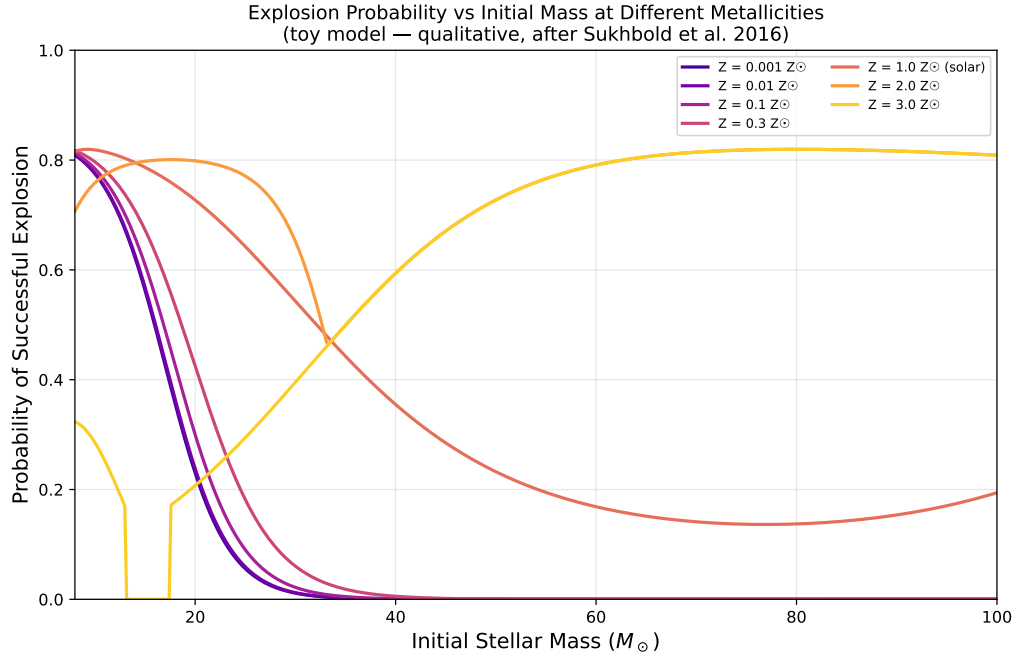


Figure 3: Explosion probability as a function of initial stellar mass for seven metallicities, computed from the toy model of Section 1.4. At solar metallicity, the probability peaks near ~ 0.85 for progenitors of $8\text{--}15 M_{\odot}$ and falls steeply above $20 M_{\odot}$ as increasing compactness makes the neutrino mechanism less effective. At low metallicity, reduced mass loss leaves more massive cores, shifting the high-probability region to lower initial masses and increasing the failed-SN fraction. The overall IMF-weighted efficiency is $\sim 0.5\text{--}0.7$ depending on metallicity (Figure 2).

3.5 Failed Supernovae: Closure Without Dissemination

When $\tau_{\text{heat}} > \tau_{\text{adv}}$ throughout the post-bounce evolution, the shock does not revive. The proto-neutron star accretes material until its mass exceeds the maximum neutron star mass (the Tolman–Oppenheimer–Volkoff limit, $M_{\text{TOV}} \approx 2\text{--}3 M_{\odot}$ depending on the nuclear equation of state Demorest2010,Cromartie2020), at which point it collapses to a black hole. The

transition from proto-neutron star to black hole occurs on a timescale of $\sim 0.5\text{--}2$ s after bounce OConnor2011.

In the language of Theorem 2.6, this is the closure case: the system fails to cross the dissemination threshold, and its contribution to interstellar medium differentiation is exactly zero. The iron, nickel, silicon, oxygen, and all other nucleosynthetic products accumulated over the star’s lifetime are enclosed within the black hole horizon. They contribute nothing to the chemical evolution of the galaxy.

The observational signature of a failed supernova is the disappearance of a massive star without a luminous transient Kochanek2008. The Large Binocular Telescope survey for failed supernovae Gerke2015,Adams2017,Neustadt2021 has identified several candidates, of which N6946-BH1 remains the most compelling: a red supergiant of initial mass $\sim 25 M_\odot$ that disappeared between 2007 and 2009 with no associated supernova Adams2017. The implied failed supernova rate from this survey is

$$f_{\text{fail}} \approx 0.14^{+0.30}_{-0.10}$$

of all stellar core collapses Neustadt2021, though the uncertainty is large and the sample is small.

This estimate has direct implications for Corollary 2.7. If $f_{\text{fail}} \sim 0.1\text{--}0.4$ of all core collapses produce no ejecta, the cumulative differentiation of the interstellar medium grows at 60–90% of the rate that would obtain if all collapses were successful. The missing differentiation is not merely delayed—it is permanently inaccessible.

3.6 What the Failed Supernova Rate Does Not Tell Us

The failed supernova rate f_{fail} is observationally accessible, in principle, through disappearing-star surveys. But the quantity that governs the differentiation of the interstellar medium is not f_{fail} alone. It is the *mass-weighted* and *composition-weighted* contribution of failed events, which depends on the mass distribution of failing progenitors.

If failures are concentrated among the most massive stars—those with the highest compactness—then the failed events are precisely the progenitors with the largest iron cores and the highest potential nucleosynthetic yield. A failed $40 M_\odot$ star carries far more nucleosynthetic differentiation behind its event horizon than a failed $10 M_\odot$ star. The composition-weighted loss is therefore not simply proportional to f_{fail} ; it depends on the *which* stars fail, not only *how many*.

The function $f(Z, M)$ in Corollary 2.8—the explosion efficiency as a function of progenitor mass and metallicity—is therefore the fundamental unknown controlling the chemical evolution

of galaxies. It cannot be determined from disappearing-star surveys alone, because those surveys are magnitude-limited and biased toward the most luminous (and most massive) progenitors. The base rate of failed explosions among stars of 8–15 $M_{\odot}$ —the dominant mass range by Salpeter weighting—is essentially unconstrained by current observations.

This is the epistemological gap of the Introduction, now stated precisely: the parameter $f(Z, M)$ is the parameter that determines how much of the chemical complexity of the universe has been irreversibly locked behind black hole horizons, and it is the parameter we know least well.

Why this section matters

The physics of core-collapse supernovae is an extended exercise in threshold dynamics. The explosion condition $\tau_{\text{heat}} < \tau_{\text{adv}}$ is the astrophysical realization of the formal dissemination threshold $E_{\text{avail}} \geq E_{\text{bind}}$. The failed supernova problem is not a curiosity at the edge of the parameter space: it is the statement that a non-negligible fraction of massive stars—possibly including the most massive and most nucleosynthetically productive—cross the closure condition rather than the dissemination threshold. The composition-weighted failed fraction $f(Z, M)$ is the central unknown governing the chemical evolution of the universe, and it is structurally inaccessible to optical transient surveys.

Self-Critique and Resolution

Concern. The explosion condition $\tau_{\text{heat}} < \tau_{\text{adv}}$ is a one-dimensional simplification. Modern 3D simulations Burrows2021, Muller2019 show that convection, SASI, and turbulent dissipation play crucial roles. Some models that fail in 1D succeed in 3D Melson2015.

Resolution. The timescale criterion illustrates the threshold condition; it does not predict outcomes quantitatively. The formal framework of Section 2 is agnostic about the physical mechanism by which the dissemination threshold is crossed; it requires only that the threshold is binary in its consequences. Whether $\tau_{\text{heat}}/\tau_{\text{adv}}$ or a 3D generalization is the correct diagnostic does not affect Theorem 2.6. *This is why we write this paper:* the binary structure of the threshold is independent of the dimensionality of the explosion model. The epistemological gap — that failed events leave no chemical trace — is a consequence of the binary structure, not of the 1D approximation.

Concern. The failed supernova rate $f_{\text{fail}} \approx 0.14^{+0.30}_{-0.10}$ Neustadt2021 is consistent with zero at 1σ . Perhaps failed supernovae are negligible for chemical evolution.

Resolution. The argument does not require a large failed fraction. Even if f_{fail} is as low as 5%, the composition-weighted contribution to lost differentiation may be significant

if failures are concentrated among the most massive progenitors — which is the physically expected case, since high-compactness progenitors are both the most likely to fail and the most nucleosynthetically productive. *This is why we write this paper*: the central claim is not that failed supernovae are frequent. It is that their frequency is unknown in precisely the mass range that matters most for nucleosynthetic yield, and that this unknown propagates directly into uncertainty about the chemical evolution of the galaxy (Corollary 2.8).

4 The Neutrino as a Structural Solution to the Closure Problem

The laws of physics do not mandate explosions. Gravity attracts without exception. Thermodynamics drives systems toward equilibrium. The second law of thermodynamics pushes toward uniformity, not differentiation. A universe written from first principles alone—gravity, thermodynamics, the tendency of bound systems toward their lowest energy state—is a universe of progressive closure: matter falling inward, gradients erasing, differentiation collapsing into homogeneity. Explosions are not written into this picture. They are, in a precise sense, *counter-natural*: local and temporary victories over the three dominant tendencies of physical systems.

And yet the universe contains supernovae. This section asks not merely how they occur, but why the physical structure of this universe contains a mechanism capable of producing them at all—and what would happen in a universe where that mechanism did not exist.

4.1 The Closure Problem at Nuclear Density

At the moment of core bounce, the collapsing iron core reaches nuclear saturation density, $\rho_{\text{nuc}} \approx 2.7 \times 10^{14} \text{ g cm}^{-3}$. At this density, matter is not merely opaque—it is, for all practical purposes of energy transport, a sealed container. Photons, which carry energy through normal stellar interiors, are trapped: the photon mean free path at nuclear density is of order femtometers, negligible compared to any macroscopic scale. Ordinary electromagnetic radiation cannot exit the core. The system is closed to light.

It is also, at first inspection, closed to everything else. The gravitational binding energy of the collapsing core is

$$E_{\text{grav}} \approx \frac{3GM_{\text{core}}^2}{5R_{\text{ns}}} \sim 3 \times 10^{53} \text{ erg},$$

of which the shock wave carries only $\sim 10^{51} \text{ erg}$ —a mere 0.3% of the available energy. The remaining 99.7% is, by all naïve expectation, inaccessible: locked inside a system so dense that no known form of matter-energy coupling can extract it on the timescale of the explosion.

This is the closure problem in its sharpest form. The energy required to explode the star exists. It was just released by the collapse of the iron core. But it is trapped inside a region so extreme that the universe’s normal mechanisms of energy transport—radiation, conduction, convection—are all inoperative. Without an additional mechanism, the collapse produces nothing but a black hole.

4.2 A Universe Without Neutrinos

Consider a hypothetical universe identical to ours in all respects except one: the neutrino does not exist, or the weak nuclear force has a cross-section so small as to be physically irrelevant on stellar scales. What happens when a massive star reaches core collapse in this universe?

The answer is unambiguous. Without a mechanism to transport energy from the collapsing core to the stalled shock, the shock cannot be revived. The neutrino heating term $\dot{Q}_\nu$ in the explosion condition $\tau_{\text{heat}} < \tau_{\text{adv}}$ (Section 3.3) vanishes identically: $\dot{Q}_\nu = 0$. The condition is never satisfied. Every core collapse is a failed supernova.

In this universe, Theorem 2.6(i) applies universally and without exception: every massive star above the Chandrasekhar mass crosses the closure threshold, collapses, and contributes exactly zero to the differentiation of the interstellar medium. The nucleosynthetic archive accumulated over millions of years of stellar evolution—the iron, the nickel, the silicon, the oxygen—disappears behind event horizons at the rate of one per century per galaxy, every century, forever.

The interstellar medium of this neutrino-free universe retains its primordial composition. Hydrogen and helium, with trace amounts of lithium from Big Bang nucleosynthesis, and nothing more. No carbon. No oxygen. No iron. No rocky planets. No chemistry beyond the simplest diatomics. The universe accumulates differentiation inside its stars and destroys it completely upon their deaths. It is, in the language of Proposition 2.9, a universe in which $\bar{f} = 0$ —and therefore a universe in which $\mathcal{D}_{\text{ISM}}(t) = \mathcal{D}_{\text{ISM}}(0)$ for all time.

This thought experiment reveals what is at stake in the neutrino mechanism. It is not merely a detail of supernova physics. It is the physical realization—in this universe, with this particle content—of the dissemination threshold itself. Remove the neutrino, and the threshold vanishes. The universe becomes chemically static.

4.3 Why Matter Must Be Mostly Empty

The neutrino solves the closure problem because of a structural feature of matter that is easy to overlook: *matter is almost entirely empty space, even at extreme densities.*

At nuclear saturation density—the densest state of matter that exists outside black holes—nucleons are separated by distances large compared to the range of the weak interaction. The weak force operates at scales of $\sim 10^{-18}$ m, set by the mass of the W and Z bosons. A nucleon has a weak-interaction cross section of

$$\sigma_\nu \sim G_F^2 E_\nu^2 \approx 10^{-41} \left(\frac{E_\nu}{10 \text{ MeV}} \right)^2 \text{ cm}^2,$$

where G_F is the Fermi coupling constant. This cross-section is extraordinarily small: a neutrino of 10 MeV energy sees a proton as a target of radius $\sim 10^{-20}$ cm—smaller than the proton itself by many orders of magnitude.

Even at nuclear density, the mean free path of a neutrino is of order

$$\lambda_\nu = \frac{1}{n \sigma_\nu} \sim 10 \left(\frac{10 \text{ MeV}}{E_\nu} \right)^2 \text{ cm},$$

where $n \sim 10^{38} \text{ cm}^{-3}$ is the nucleon number density at nuclear saturation. Ten centimeters—comparable to the radius of the proto-neutron star itself. Neutrinos are not freely streaming through the core, but they are not trapped on macroscopic timescales either. They diffuse outward on a timescale of seconds, carrying the gravitational binding energy of the neutron star through matter that is impenetrable to everything else.

This is only possible because of the quantum structure of matter. Nucleons are not continuous fluid; they are point-like particles separated by distances vastly larger than their weak-interaction cross sections. The electromagnetic cross-section of a nucleon—the scale relevant for photon transport—is of order the classical electron radius squared, $\sim 10^{-25} \text{ cm}^2$. The weak cross-section is sixteen orders of magnitude smaller. This hierarchy is what makes neutrino transport possible: the same matter that is opaque to light is nearly transparent to neutrinos.

The physical picture is stark. At nuclear density, a cubic centimeter of neutron star material contains $\sim 10^{38}$ nucleons. From the perspective of a photon, this is an impenetrable wall. From the perspective of a 10 MeV neutrino, it is a nearly empty room, with particles separated by distances that are, in units of the weak interaction range, astronomical.

The connection to the dissemination threshold is direct. The proto-neutron star is, in the formal language of Section 2, a maximally closed system: its gravitational binding energy vastly exceeds the kinetic energy of any ejecta the shock alone can produce, and it is opaque to all conventional energy transport. The only reason it is not perfectly closed—the only reason E_{avail} can exceed E_{bind} for the stellar envelope—is that neutrinos carry energy through the opaque core and deposit a fraction of it in the gain region above the shock.

In other words: the dissemination threshold is crossed, when it is crossed, because the quantum structure of matter provides a channel—the weak interaction—that is invisible to the closure mechanism. Gravity closes the system to everything it can close. The weak force opens a channel that gravity cannot close, because gravity couples to mass-energy and neutrinos, while massive, are so weakly interacting that their effective gravitational trapping radius is smaller than the proto-neutron star radius only for energies above $\sim 100 \text{ MeV}$ —far above the thermal energies relevant here [Bethe1985](#), [Janka2012](#).

4.4 The Physical Basis of the Selection Argument

This analysis reframes the persistence argument of Section ?? in physical terms. The question is not merely whether disseminating systems exist—by observation they do—but *why this universe contains a physical mechanism capable of crossing the closure threshold at all*.

The answer has three layers.

Layer 1: The hierarchy of forces. The Standard Model of particle physics contains four fundamental interactions with vastly different coupling strengths. The weak force is weaker than electromagnetism by many orders of magnitude at low energies, which is why neutrino cross-sections are so small. This hierarchy is a contingent feature of this universe—there is no known principle requiring it. In a universe where the weak force had electromagnetic-scale coupling, neutrinos would be trapped in the core as efficiently as photons, and the neutrino mechanism would not function.

Layer 2: The structure of matter. Matter is composed of quarks and leptons—point-like, or very nearly so—interacting through fields rather than through contact. This means that even at extreme densities, most of the volume of matter is dynamically empty from the perspective of weak interactions. A universe in which matter were a true continuum at nuclear scales would have no weak transparency, regardless of the coupling constant.

Layer 3: The energy scale of collapse. The gravitational energy released in core collapse produces neutrinos at thermal energies of $\sim 10\text{--}20$ MeV. At these energies, the neutrino mean free path through the gain region is of order 10^7 cm—comparable to the gain region depth—which is precisely the scale at which neutrino heating is effective: not so large that neutrinos stream freely without depositing energy, not so small that they are trapped without reaching the shock. This coincidence of scales—collapse energy, neutrino cross-section, and gain region geometry—is what makes the mechanism work at the quantitative level Janka2001,Burrows2021.

None of these three layers was designed to produce supernovae. But together they constitute a physical configuration in which the closure problem has a solution. The neutrino is the universe’s answer to a question the universe did not ask: how to carry energy through matter that is opaque to everything else.

Why this section matters

Explosions are not written into the laws of physics. They are counter-natural: local victories over gravity, thermodynamics, and the inertia of bound systems. The reason core-collapse supernovae exist at all—the reason the dissemination threshold can be crossed in this universe—is that the quantum structure of matter provides a transport channel, the weak interaction, that is invisible to the closure mechanism. A universe without neutrinos, or with a weak force too weak to operate on stellar scales, would have $\bar{f} = 0$: every core collapse would be a failed supernova, the interstellar medium would retain its primordial composition forever, and the chemical complexity we observe would not exist. The neutrino is not a detail of supernova physics. It is the physical realization of the dissemination threshold in this universe.

Self-Critique and Resolution

Concern. The thought experiment of a neutrino-free universe proves nothing about the actual universe. The neutrino exists. What does the counterfactual add?

Resolution. The counterfactual is an isolation argument: it identifies precisely which physical ingredient is responsible for crossing the dissemination threshold. By removing it and tracing the consequences, we establish that successful supernovae in this universe are not a generic consequence of gravitational physics but a specific consequence of the particle content and force hierarchy of this universe. *This is why we write this paper:* the existence of the dissemination mechanism is itself a contingent and remarkable fact. The field of chemical evolution has treated the explosion rate as a parameter to be measured rather than a phenomenon to be explained at the level of physical principles. The neutrino section provides that explanation. Explosions are counter-natural. They persist because this universe happens to contain a transport channel — the weak interaction — that gravity cannot close.

Concern. Type Ia supernovae do not involve neutrinos. Does the argument apply to them?

Resolution. Yes, structurally, through a different physical channel. In Type Ia supernovae, the closure problem is solved by thermonuclear energy: carbon burning releases energy exceeding the white dwarf's gravitational binding by a factor of ~ 3 (Section 5.1), and the deflagration-to-detonation transition provides rapid energy release that overcomes binding. The physical mechanism is different — strong-force energy release rather than weak-force energy transport — but the structural role is identical: a channel that the closure mechanism cannot suppress. *This is why we write this paper:* Theorem 2.6 encompasses both channels through the same formal structure. The diversity of physical mechanisms that realize the

dissemination threshold is itself evidence for the selectional argument: wherever differentiation must be disseminated, physics finds a way — not by design, but because systems that do not find a way leave no record.

5 Type Ia Supernovae

Type Ia supernovae occupy a structurally distinct position in the dissemination landscape. Unlike core-collapse supernovae, they involve no iron-core formation, no neutron star, and no neutrino-driven mechanism. The energy source is thermonuclear: the explosive burning of a carbon-oxygen white dwarf that has approached or reached the Chandrasekhar mass. The explosion, when it occurs, releases $\sim 10^{51}$ erg of kinetic energy and synthesizes $\sim 0.5\text{--}1.0 M_\odot$ of ^{56}Ni , whose radioactive decay powers the optical light curve. The peak luminosity of this decay chain is sufficiently uniform—after empirical standardization—to serve as a cosmological distance indicator Phillips1993,Riess1998,Perlmutter1999.

But the uniformity of the *successful* explosion conceals a prior question that is rarely foregrounded: what fraction of the systems that could in principle produce a Type Ia supernova actually do so? This section examines Type Ia supernovae as dissemination events in the formal sense of Section 2, and shows that the unknown detonation efficiency introduces a systematic uncertainty into the use of Type Ia as standard candles that is epistemologically distinct from, and potentially more fundamental than, the photometric uncertainties that have been extensively studied.

5.1 The White Dwarf as a Gravitationally Closed System

A carbon-oxygen white dwarf is, in the language of Definition 2.4, a closed system. Its differentiated content—the result of helium burning and subsequent carbon-oxygen stratification in the progenitor AGB star—is gravitationally confined by electron degeneracy pressure. The binding energy of a white dwarf near the Chandrasekhar mass is

$$E_{\text{bind}}^{\text{WD}} \approx \frac{3}{5} \frac{GM_{\text{Ch}}^2}{R_{\text{WD}}} \approx 5 \times 10^{50} \text{ erg},$$

where $R_{\text{WD}} \approx 3000$ km for a near-Chandrasekhar white dwarf Shapiro1983. This is substantially less than the nuclear energy available from complete thermonuclear burning of the white dwarf material:

$$E_{\text{nuc}} = \Delta\epsilon_{\text{nuc}} M_{\text{Ch}} \approx 1.5 \times 10^{51} \text{ erg},$$

where $\Delta\epsilon_{\text{nuc}} \approx 7 \times 10^{17}$ erg g $^{-1}$ is the energy released per unit mass in the carbon-to-nickel burning chain Arnett1996. The thermonuclear energy therefore exceeds the gravitational binding energy by a factor of ~ 3 : in principle, complete burning provides sufficient energy to unbind the white dwarf entirely and cross the dissemination threshold.

The crucial word is *in principle*. Whether the nuclear energy is actually released as an

explosion—rather than being consumed by other processes or released gradually—depends on the ignition and propagation of the thermonuclear flame, and this is where the uncertainty lies.

5.2 Ignition, Deflagration, and Detonation

As the white dwarf accretes matter from a companion and its mass approaches M_{Ch} , the central density and temperature rise. Carbon ignites at the center when the temperature reaches $\sim 7 \times 10^8$ K at densities of $\sim 2 \times 10^9$ g cm $^{-3}$ Nomoto1984. At these densities, the burning front propagates initially as a subsonic deflagration (a flame), not a supersonic detonation (a shock-mediated burning front). The deflagration–detonation transition (DDT) is the central unresolved problem of Type Ia progenitor theory.

A pure deflagration releases sufficient energy to pre-expand the white dwarf, reducing the density before the flame can burn the entire mass. If the deflagration does not transition to a detonation, the result is a sub-energetic, asymmetric explosion that produces too little ^{56}Ni and leaves behind a partially burned remnant Roepke2007. This outcome crosses the dissemination threshold in the sense that some ejecta are produced, but the nucleosynthetic yield is reduced.

A successful DDT—where the deflagration transitions to a detonation in the turbulent, pre-expanded white dwarf—burns the remaining material at high density and produces the canonical $\sim 0.6 M_{\odot}$ of ^{56}Ni observed in normal Type Ia supernovae Iwamoto1999. This is the scenario implicitly assumed by the standard-candle method.

A third outcome is also possible: accretion-induced collapse (AIC), in which the white dwarf collapses to a neutron star rather than detonating Nomoto1991,Dessart2006. This occurs when the central density reaches $\sim 4 \times 10^9$ g cm $^{-3}$ before carbon ignition is vigorous enough to launch a deflagration. The collapse is rapid, the result is a neutron star, and no thermonuclear ejecta are produced. In the language of Theorem 2.6(i), this is the zero-contribution case: a system that approaches the dissemination threshold and fails to cross it.

The relative probabilities of these three outcomes—successful DDT, sub-energetic deflagration, and AIC—depend on the white dwarf central density at ignition, the ignition geometry, the turbulent flame structure, and the metallicity of the progenitor Hillebrandt2013,Maoz2014. None of these probabilities is directly observable.

5.3 The Unknown Detonation Efficiency

The Type Ia detonation efficiency η_{Ia} is defined here as the fraction of systems that reach near-Chandrasekhar white dwarf mass and produce a successful thermonuclear explosion with canonical nucleosynthetic yield. This parameter is analogous to the explosion efficiency $f(Z, M)$ of the core-collapse case: it quantifies the fraction of systems that cross the dissemination threshold rather than failing into the closure case.

The observational constraints on η_{Ia} are indirect. The Type Ia rate per unit stellar mass—the specific rate—is observed to be $\sim 1\text{--}3 \times 10^{-3}$ SNe per solar mass of formed stars per Gyr in late-type galaxies Maoz2012. This rate can be compared to theoretical predictions for the number of systems that reach the Chandrasekhar mass in a given stellar population. However, this comparison requires knowledge of the binary fraction, mass-transfer efficiency, and donor star mass function, none of which are precisely known Ruiter2009, Toonen2012.

The result is that η_{Ia} is constrained only within a factor of several. Estimates range from $\sim 1\%$ to $\sim 10\%$ of all white dwarfs in appropriate binary systems eventually producing Type Ia explosions Maoz2014. The remaining 90–99% either undergo AIC (zero contribution), produce sub-luminous deflagrations (partial contribution), or are disrupted before reaching the Chandrasekhar mass. This wide range reflects genuine physical uncertainty, not merely observational limitation.

5.4 Implications for Cosmological Distance Measurements

The use of Type Ia supernovae as cosmological standard candles rests on two empirical facts: the peak luminosity of Type Ia events correlates with the light-curve decline rate Phillips1993, and this correlation can be calibrated to reduce the intrinsic scatter to ~ 0.15 mag Guy2007. The discovery of the accelerating expansion of the universe by Riess1998 and Perlmutter1999 was built on this calibration.

The standard-candle method implicitly assumes that the observed sample of Type Ia supernovae is representative of the underlying population of Chandrasekhar-mass thermonuclear events. If η_{Ia} is redshift-dependent—if the fraction of systems that successfully detonate varies with cosmic time—then the observed sample at different redshifts is drawn from populations with different explosion efficiencies, and the standardization may be systematically biased.

Several physical mechanisms could produce a redshift-dependent η_{Ia} . Metallicity evolves with redshift Mannucci2010: the progenitor white dwarf metallicity affects the central carbon-to-oxygen ratio at ignition Timmes2003, which in turn affects the ^{56}Ni yield and the deflagration-to-detonation transition probability Krueger2010. The binary star fraction and orbital period distribution evolve with the star formation history Moe2017. The delay time

distribution—the time between star formation and Type Ia explosion—convolved with the star formation rate produces a progenitor population whose properties change with redshift Graur2013,Maoz2012.

These effects are not captured by the empirical light-curve standardization, which corrects for observed photometric variations but not for systematic shifts in the underlying detonation physics. The formal expression of this uncertainty follows from Corollary 2.8: the rate of ISM differentiation from Type Ia events at redshift z is

$$\left. \frac{d\mathcal{D}_{\text{ISM}}}{dt} \right|_{\text{Ia}}(z) = \dot{n}_{\text{Ia}}(z) \eta_{\text{Ia}}(z) \Delta\mathcal{D}_{\text{Ia}},$$

where $\dot{n}_{\text{Ia}}(z)$ is the rate of Chandrasekhar-mass events, $\eta_{\text{Ia}}(z)$ is the redshift-dependent detonation efficiency, and $\Delta\mathcal{D}_{\text{Ia}}$ is the differentiation contribution per successful event. The cosmological distance measurement is sensitive to $\dot{n}_{\text{Ia}}(z)$ through the luminosity distance; it is not sensitive to $\eta_{\text{Ia}}(z)$ because failed events (AIC, sub-energetic deflagrations) are not included in the Type Ia sample. However, if $\eta_{\text{Ia}}(z)$ varies systematically with z , the surviving sample at different redshifts may differ in ways that the photometric standardization does not fully correct.

We do not claim that this effect is large enough to alter cosmological conclusions. We claim that it is *unquantified*, because $\eta_{\text{Ia}}(z)$ is not observationally accessible—accretion-induced collapses leave no Type Ia-like optical transient, and their rate relative to successful Type Ia events is unknown at any redshift. This is the Type Ia analog of the core-collapse epistemological gap: the failed events are invisible, and their frequency is unmeasured.

5.5 A Toy Model for Detonation Efficiency

To illustrate the potential magnitude of the effect, we construct a minimal model for the redshift evolution of $\eta_{\text{Ia}}(z)$. We parameterize the detonation efficiency as a function of the mean progenitor metallicity $\langle Z \rangle(z)$, using the observed mass-metallicity relation evolved to higher redshift Mannucci2010:

$$\langle Z \rangle(z) = Z_0 \exp\left(-\frac{z}{z_0}\right),$$

where $Z_0 \approx Z_{\odot}$ is the local mean metallicity and $z_0 \approx 2$ is a characteristic enrichment redshift. The detonation efficiency is modeled as

$$\eta_{\text{Ia}}(z) = \eta_0 \left[1 + \alpha_Z \frac{\langle Z \rangle(z) - Z_0}{Z_0} \right],$$

where η_0 is the local efficiency and α_Z parameterizes the metallicity sensitivity. Figure 4 shows $\eta_{\text{Ia}}(z)$ for three values of $\alpha_Z \in \{-0.3, 0, +0.3\}$, spanning the plausible range of metallicity sensitivity Krueger2010,Timmes2003.

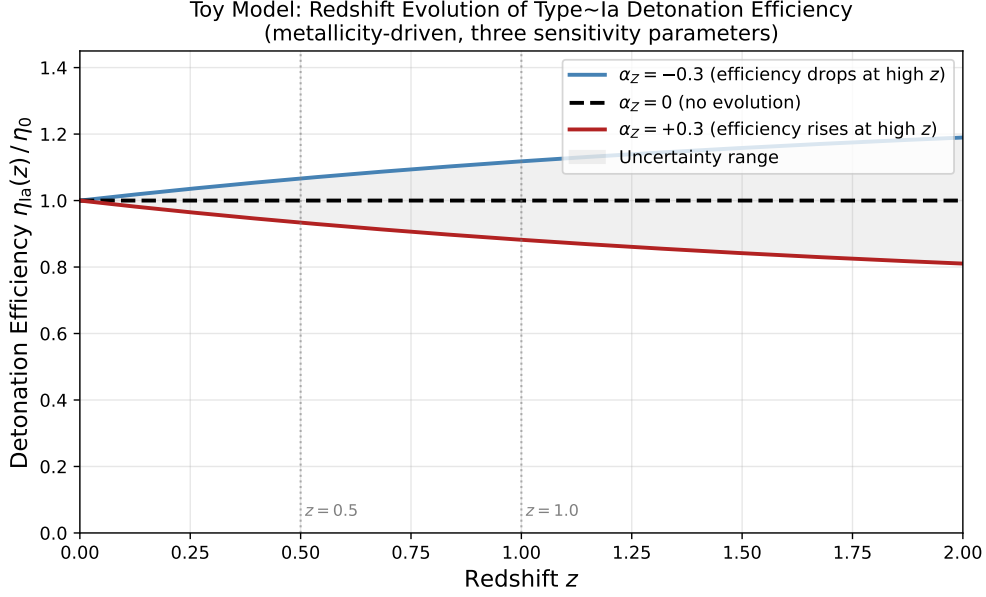


Figure 4: Toy model for the redshift evolution of the Type Ia detonation efficiency $\eta_{\text{Ia}}(z)$, normalized to the local value η_0 . Three curves show different metallicity sensitivities α_Z : negative (efficiency decreasing with redshift, blue), zero (no evolution, black dashed), and positive (efficiency increasing with redshift, red). The shaded region illustrates the uncertainty range. If $\eta_{\text{Ia}}(z)$ is not constant, the surviving Type Ia sample at different redshifts is drawn from populations with different intrinsic properties, introducing a systematic that is not corrected by photometric standardization.

The point of this model is not to predict the sign or magnitude of α_Z —that would require detailed explosion simulations at multiple metallicities, which are beyond the scope of this paper Seitzzahl2013,Fink2014. The point is to show that even a modest metallicity dependence ($|\alpha_Z| \sim 0.3$) produces a variation in η_{Ia} of $\sim 15\text{--}20\%$ between $z = 0$ and $z = 1$, the redshift range over which the accelerating expansion was discovered. Whether this variation affects cosmological conclusions depends on how strongly η_{Ia} correlates with the observable light-curve properties used for standardization—a correlation that has not been established, because the failed events that would define one end of the distribution are not observed.

Why this section matters

Type Ia supernovae are the primary tool of observational cosmology over the redshift range $0 < z < 2$. The standard-candle method assumes that the sample of observed events is representative of the underlying thermonuclear explosion population. But the fraction of near-Chandrasekhar white dwarfs that successfully detonate—rather than collapsing to neutron stars or producing sub-energetic deflagrations—is not observationally constrained, and may evolve with redshift through metallicity and binary fraction effects. This is a direct application of Theorem 2.6: the closed systems (AIC events, failed deflagrations) leave no luminous trace and cannot be counted. Their unknown frequency introduces an irreducible systematic into the Type Ia distance scale that is epistemologically distinct from photometric uncertainties.

Self-Critique and Resolution

Concern. The accretion-induced collapse channel may be rare. Some theoretical models suggest that the parameter space for AIC is narrow and that most near-Chandrasekhar white dwarfs either detonate or disrupt without collapsing Dessart2006,Ruiter2019.

Resolution. The argument does not require AIC to be common. It requires only that the fraction of near-Chandrasekhar systems that do not produce canonical Type Ia events is nonzero and potentially redshift-dependent. Even if AIC contributes only a few percent of all Chandrasekhar-mass events, the systematic shift in the surviving sample may be detectable with next-generation surveys LSST2009, Euclid2011. *This is why we write this paper:* the undetectable fraction — however small — is the fraction whose chemical contribution is exactly zero. Its existence means that the Type Ia sample is not a complete census of thermonuclear events, and that the standard-candle method rests on a sample whose selection function is unknown at the level of the detonation efficiency.

Concern. Photometric standardization already corrects for observed variations in light-curve shape, color, and host-galaxy properties. Why would residuals retain a dependence on $\eta_{\text{Ia}}(z)$?

Resolution. The standardization corrects for properties of the *observed* sample. Failed events (AIC, sub-energetic deflagrations) that would occupy a different region of the light-curve parameter space are removed from the sample before standardization is applied. The bias is proportional to the covariance between η_{Ia} and the standardizable light-curve parameters — a quantity that cannot be measured from the observed sample alone, because the events that would define one end of the distribution are not observed. *This is why we write this paper:* the epistemological structure of the Type Ia problem is identical to that of the core-collapse

problem. In both cases, the systems that fail to cross the dissemination threshold are absent from any survey that detects their emission. The unknown is not in the data. It is in what the data does not contain.

6 Epistemological Implications and the Fragility of Chemical Evolution

The preceding sections have established two parallel results. In core-collapse supernovae, the fraction of massive stars that successfully explode—rather than collapsing directly to black holes—is a poorly constrained function of progenitor mass and metallicity, and failed events are chemically invisible to optical surveys. In Type Ia supernovae, the fraction of near-Chandrasekhar white dwarfs that detonate rather than undergoing accretion-induced collapse is similarly unconstrained, and may evolve with redshift in ways that standard-candle photometry cannot detect.

In both cases, the formal framework of Section 2 identifies the same structure: the growth of interstellar medium differentiation depends on a parameter—the explosion efficiency—that is not directly observable because the systems that fail to cross the dissemination threshold leave no chemical trace. This section examines the consequences of this structure for astrophysics as an epistemological enterprise, and for the question of whether the chemical complexity of the universe is robust or fragile.

6.1 The Structural Inaccessibility of the Failed-Event Base Rate

Standard observational astronomy is, by construction, a science of emitters. A telescope detects photons; a gravitational wave detector detects metric perturbations; a neutrino detector detects flavor-converted leptons. All of these signals require a source. A system that collapses to a black hole without a successful shock, or that undergoes accretion-induced collapse without a thermonuclear runaway, produces none of these signals at chemically significant levels. It is, in the observationally relevant sense, absent.

This is not a contingent technological limitation. It is a structural feature of the measurement problem. To determine the failed-event base rate f_{fail} , one must count both successes and failures. Successes are counted by detecting their emissions. Failures must be counted by detecting the *absence* of an expected emission—which requires knowing, in advance, which systems were expected to emit. This is the disappearing-star approach of Kochanek2008 and the LBT survey Gerke2015,Adams2017,Neustadt2021: monitor a large sample of massive stars and record those that vanish without a supernova.

This approach is powerful but limited in three ways that are intrinsic to the method, not to the current state of instrumentation.

First, it requires a prior catalog of potential progenitors. The catalog is magnitude-limited: only stars bright enough to be resolved in the monitoring survey can be included. For galaxies

beyond the Local Group, individual stellar resolution is not possible, and the failed-SN rate in the mass range 8–15 $M_{\odot}$ —the dominant range by Salpeter weighting—is inaccessible at any practical survey depth.

Second, the disappearance of a star is not an unambiguous signal of failed collapse. The star may have been obscured by dust, may have evolved to a configuration below the survey detection threshold, or may have been disrupted in a binary interaction Kochanek2008,Reynolds2015. Each candidate requires extensive follow-up, limiting the sample to a handful of events per decade even with dedicated monitoring.

Third, and most fundamentally, the method measures the rate of optically disappearing stars, not the rate of chemically failed collapses. A failed collapse that produces a faint transient from the pre-collapse wind Lovegrove2013 or a brief neutrino burst Beacom2010 may not appear as a disappearing star in an optical survey, yet it contributes zero to ISM differentiation. The chemical failure rate and the optical disappearance rate are related but not identical.

The conclusion is that the parameter $f(Z, M)$ in Corollary 2.8 is structurally inaccessible to any survey that relies on electromagnetic emission. Next-generation gravitational wave detectors may detect failed collapses through their metric signature Gossan2016,Ott2009, but the chemical contribution of a gravitationally detected failed collapse is still zero. The gravitational wave measurement counts failures; it does not undo their chemical consequence.

6.2 The Fragility Argument

The persistence argument of Section 1 established that the existence of a chemically differentiated ISM requires that the explosion efficiency be positive. It does not establish that the efficiency is large, or that it is stable across cosmic time. We now ask the sharper question: how sensitive is the chemical evolution of the universe to the value of the explosion efficiency?

From Corollary 2.8, the total ISM differentiation after n stellar generations is proportional to the cumulative number of successful dissemination events. If the explosion efficiency $\bar{f}$ is reduced by a factor $\lambda < 1$, the rate of ISM enrichment is reduced by the same factor, and the time required to reach the present level of ISM differentiation is increased by $1/\lambda$.

This scaling has a concrete implication. The solar system formed approximately 4.6 Gyr ago from material enriched by several generations of massive star evolution Lodders2003. The iron abundance of the proto-solar nebula was $[\text{Fe}/\text{H}] \approx 0$, placing it at the median of the local disk metallicity distribution Casagrande2011. This level of enrichment required a cumulative number of successful supernova explosions estimable from galactic chemical

evolution models Kobayashi2006:

$$N_{\text{SN}}^{\text{req}} \sim 10^9\text{--}10^{10} \quad (\text{Milky Way disk, 10 Gyr of evolution}).$$

If the actual explosion efficiency $\bar{f}$ is a factor of two lower than assumed in standard chemical evolution models, the Milky Way disk would require twice as long to reach solar metallicity—approximately 20 Gyr, exceeding the age of the universe by a factor of ~ 1.5 . A solar system forming under these conditions would not exist at the present cosmic epoch.

This is not a prediction that the efficiency is low. It is a demonstration that the chemical history of the galaxy is sensitive to the explosion efficiency at the factor-of-two level, and that the efficiency is not known to better than a factor of several. The fragility of chemical evolution is therefore not a theoretical conjecture; it is a direct consequence of the epistemological gap.

6.3 A Formal Fragility Measure

Definition 6.1 (Chemical fragility). *Let $\mathcal{D}_{\text{ISM}}(t; \bar{f})$ denote the ISM differentiation at time t under explosion efficiency $\bar{f}$. The chemical fragility of the ISM at time t is*

$$\mathcal{F}(t) = - \frac{\partial \ln \mathcal{D}_{\text{ISM}}}{\partial \ln \bar{f}} \bigg|_{\bar{f}=\bar{f}_0},$$

where $\bar{f}_0$ is the fiducial efficiency. A fragility of $\mathcal{F} = 1$ means that a 10% reduction in explosion efficiency produces a 10% reduction in ISM differentiation at time t .

Proposition 6.2 (Linear fragility in the closed-box regime). *In the linear enrichment regime (ISM mass much larger than ejecta mass per event, no re-enrichment of ejecta sources, primordial initial conditions), $\mathcal{F}(t) = 1$ for all t .*

Proof. From Corollary 2.7, $\mathcal{D}_{\text{ISM}}^{(n)} = \mathcal{D}_{\text{ISM}}^{(0)} + \bar{f} n \langle \Delta \mathcal{D} \rangle$, where $\langle \Delta \mathcal{D} \rangle$ is the mean differentiation contribution per successful event and n is the total number of potential events. Differentiating:

$$\frac{\partial \mathcal{D}_{\text{ISM}}}{\partial \bar{f}} = n \langle \Delta \mathcal{D} \rangle.$$

In the limit $\mathcal{D}_{\text{ISM}}^{(0)} \rightarrow 0$ and large n :

$$\mathcal{F} = - \frac{\bar{f}}{\mathcal{D}_{\text{ISM}}} \cdot n \langle \Delta \mathcal{D} \rangle \rightarrow 1. \quad \square$$

□

Linear fragility means that the uncertainty in $\bar{f}$ propagates directly and without amplification into uncertainty in $\mathcal{D}_{\text{ISM}}$. If $\bar{f}$ is uncertain by a factor of two—consistent with the observational constraints of Sections 3 and 5—then $\mathcal{D}_{\text{ISM}}(t)$ is uncertain by the same factor at any given time. This is a large uncertainty for a quantity that governs the formation of rocky planets and the preconditions for biological complexity.

6.4 Observable Consequences: Chemical Evolution Rates

If the explosion efficiency $\bar{f}(z)$ varies with redshift, the rate of chemical evolution of galaxies will show a redshift dependence not fully captured by the observed star formation rate. In simple closed-box chemical evolution Pagel1997, the predicted ISM iron abundance at redshift z is:

$$[\text{Fe}/\text{H}](z) = \ln \left[1 + \bar{f}(z) y_{\text{Fe}} \frac{\mu_{\star}(z)}{\mu_g(z)} \right],$$

where y_{Fe} is the iron yield per unit mass of stars formed, $\mu_{\star}$ is the stellar mass fraction, and μ_g is the gas fraction. If $\bar{f}(z)$ is redshift-dependent, the observed mass-metallicity relation Mannucci2010,Maiolino2019 will deviate from predictions of models that assume constant $\bar{f}$.

Figure 5 illustrates the predicted $[\text{Fe}/\text{H}]$ evolution for three efficiency scenarios. A factor-of-two variation in $\bar{f}$ between $z = 0$ and $z = 2$ produces a metallicity difference of ~ 0.2 – 0.3 dex at fixed stellar mass—comparable to the current observational scatter in the mass-metallicity relation Mannucci2010,Curti2020. The uncertainty in $\bar{f}(z)$ is therefore not negligible relative to the observational precision of current metallicity surveys.

6.5 Summary of Principal Unknowns

We collect the principal unknowns identified in this paper and their consequences.

The core-collapse explosion efficiency $f(Z, M)$. Constrained by disappearing-star surveys to $f_{\text{fail}} \approx 0.14^{+0.30}_{-0.10}$ Neustadt2021, but only for luminous progenitors in nearby galaxies. The efficiency in the dominant Salpeter mass range (8–15 $M_{\odot}$) is essentially unconstrained. The full function $f(Z, M)$ cannot be determined from current observations.

The Type Ia detonation efficiency $\eta_{\text{Ia}}(z)$. Constrained only within a factor of several from binary population synthesis models Ruiter2009,Toonen2012. Its redshift evolution is essentially unknown. Both quantities enter the chemical evolution rate of iron-group elements Matteucci1986.

The composition-weighted failed fraction. Even if the number fraction of failed events were known, the quantity governing chemical evolution is the mass-weighted and composition-weighted contribution of failing progenitors. A failed 40 $M_{\odot}$ star carries far more

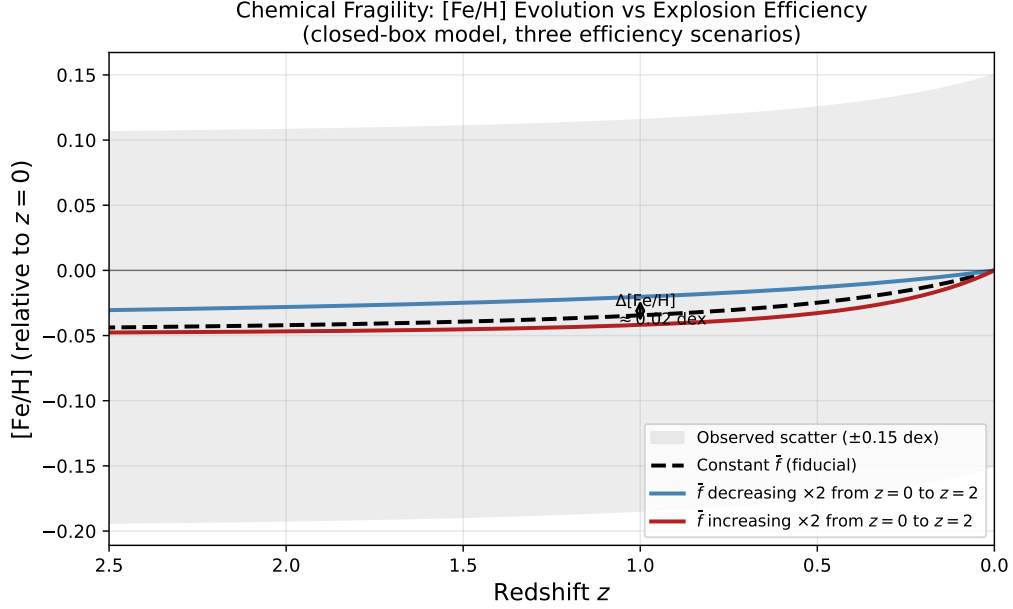


Figure 5: Predicted ISM iron abundance $[\text{Fe}/\text{H}]$ as a function of redshift for three assumptions about the evolution of the explosion efficiency $\bar{f}(z)$: constant at the local value (black dashed), decreasing by a factor of two from $z = 0$ to $z = 2$ (blue), and increasing by a factor of two (red). The shaded band shows the approximate 1σ scatter in the observed mass-metallicity relation Mannucci2010. A factor-of-two variation in $\bar{f}$ produces a shift of $\sim 0.2\text{--}0.3$ dex, comparable to this scatter. The explosion efficiency is therefore an unmodeled systematic in current chemical evolution studies.

potential nucleosynthesis behind its event horizon than a failed $10 M_{\odot}$ star. This quantity cannot be inferred from any current observational program.

The time evolution of $\bar{f}$. Both $f(Z, M)$ and η_{Ia} depend on metallicity, which evolves with cosmic time. The explosion efficiency therefore has a cosmic history coupled to the chemical evolution it drives, creating a non-linear feedback: higher efficiency produces faster enrichment, higher metallicity, and changed efficiency for the next stellar generation. The fixed points and stability of this coupled system have not been systematically analyzed.

Why this section matters

The epistemological gap—the structural inaccessibility of the failed-event base rate—follows from the binary structure of Theorem 2.6: systems that do not cross the dissemination threshold leave no chemical trace and cannot be counted by any method relying on chemical or electromagnetic emission. The linear fragility result (Proposition 6.2) shows that this unobservable parameter determines, within a factor of two, the metallicity of the ISM at any given cosmic epoch—and therefore the conditions under which planetary systems and biological chemistry can form.

Self-Critique and Resolution

Concern. The fragility argument assumes a simple linear scaling between explosion efficiency and ISM differentiation. Real galactic chemical evolution involves inflows, outflows, radial mixing, and metallicity-dependent yields Kobayashi2006,Prantzos2018. The sensitivity to $\bar{f}$ may be reduced by compensating effects.

Resolution. Proposition 6.2 proves linear fragility in the closed-box limit. In open-box models with gas inflows, inflow dilutes both ejecta iron and total gas mass equally in the steady-state limit, preserving proportionality to $\bar{f}$. Outflows driven by supernova feedback introduce a correlation between $\bar{f}$ and the outflow rate, producing superlinear fragility ($\mathcal{F} > 1$). The linear estimate is a lower bound on fragility, not an overestimate. *This is why we write this paper:* even the most conservative fragility estimate — linear, closed-box — implies that a factor-of-two uncertainty in the explosion efficiency propagates to a factor-of-two uncertainty in the metallicity of the ISM at any cosmic epoch. This is not a second-order correction. It is an uncertainty of order unity in the central variable of galactic chemical evolution, and it has not been propagated through the models that depend on it.

Concern. The factor-of-two uncertainty in $\bar{f}$ may be too pessimistic. Detailed models Sukhbold2016,Burrows2021 constrain explosion outcomes for specific progenitor structures to better precision.

Resolution. The uncertainty in $\bar{f}$ has two components: the theoretical uncertainty in the explosion outcome for a given progenitor structure (which simulations address), and the observational uncertainty in the progenitor structure itself — the distribution of rotation rates, magnetic fields, and binary histories. The second component is not reduced by more detailed explosion simulations. It requires improved pre-supernova observations, available for only a small number of events Smartt2009, VanDyk2017. The factor-of-two estimate reflects the combined uncertainty. *This is why we write this paper:* the uncertainty is not primarily theoretical. It is observational, structural, and persistent. It will not be resolved by

better explosion simulations alone. It requires confronting the epistemological gap this paper identifies: that the composition-weighted failed fraction is inaccessible to any survey that detects only the emitters.

7 Conclusions

This paper has developed and applied a formal framework for a claim that is physically elementary but rarely stated with precision: the chemical enrichment of the interstellar medium depends not on the synthesis of heavy elements inside stars, but on the subsequent explosion that releases them. Synthesis without dissemination is thermodynamically inert from the perspective of the surrounding medium. The star that spends ten million years building an iron core and then collapses directly to a black hole contributes exactly as much to the chemical evolution of the galaxy as a star that never formed at all.

We have formalized this claim in Theorem 2.6, proved it by two independent methods, and derived its consequences through two corollaries and a fragility proposition. We have applied the framework to the two main supernova channels—core-collapse and Type Ia—and identified in each channel an epistemological gap: a parameter that governs the rate of ISM enrichment but is structurally inaccessible to observational programs that rely on detecting electromagnetic emission.

We summarize the principal conclusions.

7.1 On the Formal Structure

The central theorem (Section 2) establishes that gravitationally bound systems divide into two classes with respect to their contribution to ISM differentiation: those that cross the dissemination threshold ($E_{\text{avail}} \geq E_{\text{bind}}$) and those that do not. The contribution of the first class is strictly positive; the contribution of the second class is exactly zero. There is no intermediate case.

This binary structure is not an artifact of the formal framework. It reflects a physical reality: matter inside a black hole horizon cannot contribute to the chemical state of the surrounding medium by any causal mechanism. The formalism makes explicit what is implicit in standard stellar evolution theory, and draws attention to the threshold condition as the governing variable of chemical evolution.

The formal framework is deliberately stated in terms of gravitational binding energy rather than in abstract substrate-independent language, because gravity is the actual closure mechanism in all cases of astrophysical interest. The generality of the mathematical structure—which would apply to any system with a closure mechanism and an internal differentiation—is acknowledged but not pursued here.

7.2 On Core-Collapse Supernovae

The neutrino-driven explosion condition $\tau_{\text{heat}} < \tau_{\text{adv}}$ is the astrophysical realization of the dissemination threshold. When this condition is not met, the proto-neutron star accretes to the Tolman–Oppenheimer–Volkoff limit and collapses to a black hole. The nucleosynthetic archive accumulated during the progenitor’s lifetime is permanently lost.

The fraction of core collapses that fail to meet this condition—the failed supernova fraction f_{fail} —is constrained by disappearing-star surveys to $\approx 0.14^{+0.30}_{-0.10}$ Neustadt2021, but only for luminous progenitors in nearby galaxies. The composition-weighted failed fraction, which enters the chemical evolution rate, cannot be determined from optical surveys: it depends on the mass distribution of failing progenitors, precisely the information absent when a progenitor collapses without a transient.

The explosion landscape in the (initial mass, metallicity) parameter space is non-monotonic (Section 3.4), with successful explosions concentrated at low initial mass and increasing failure probability at high mass. Metallicity modulates the landscape through mass loss, but does not eliminate the dominant failed-SN region at high initial mass at any metallicity.

7.3 On Type Ia Supernovae

Type Ia supernovae present a structurally analogous problem. The fraction of near-Chandrasekhar white dwarfs that successfully detonate—rather than undergoing accretion-induced collapse or producing sub-energetic deflagrations—is constrained only within a factor of several by binary population synthesis models Ruiter2009, Toonen2012. This detonation efficiency η_{Ia} may evolve with redshift through metallicity and binary fraction effects.

The standard-candle method used in cosmological distance measurements Riess1998, Perlmutter1999 corrects for photometric variations in the observed Type Ia sample. It does not correct for a redshift-dependent selection effect introduced by the unknown $\eta_{\text{Ia}}(z)$: if the fraction of near-Chandrasekhar events that successfully detonate varies with redshift, the surviving sample at different redshifts is drawn from populations with systematically different intrinsic properties. The magnitude of this effect, modeled qualitatively in Section 5.5, is potentially detectable with next-generation surveys.

7.4 On Chemical Fragility

The chemical fragility of the ISM—the sensitivity of its differentiation to the explosion efficiency—is linear in the closed-box regime (Proposition 6.2): a factor-of-two uncertainty in $\bar{f}$ propagates to a factor-of-two uncertainty in $\mathcal{D}_{\text{ISM}}(t)$ at any given time. In open-box

models with supernova-driven outflows, the fragility is superlinear.

A factor-of-two reduction in the explosion efficiency relative to standard chemical evolution models would require the Milky Way disk approximately 20 Gyr to reach solar metallicity—exceeding the age of the universe. This calculation is not a prediction; it is a demonstration of sensitivity. The explosion efficiency is not known to better than a factor of several, and this uncertainty has not been systematically propagated through galactic chemical evolution models.

7.5 On the Persistence Argument

The observed chemical differentiation of the universe—the existence of iron, nickel, calcium, and the other heavy elements that constitute rocky planets and biological molecules—requires that the explosion efficiency be positive (Proposition 2.9). This statement does not depend on the existence of observers and is not an anthropic argument. It follows from the definition of differentiation and the binary structure of Theorem 2.6: if no system ever crossed the dissemination threshold, the ISM would retain its primordial composition indefinitely.

This argument applies to any universe containing gravitationally bound systems capable of internal nucleosynthesis. In any such universe, the persistence of chemical differentiation across stellar generations is logically equivalent to the existence of a population of systems that successfully crossed the dissemination threshold. The chemical complexity we observe is the cumulative record of all such crossings; the systems that failed to cross left no record.

7.6 Open Questions

The framework developed here suggests several specific observational and theoretical questions that have not been systematically addressed.

What is the composition-weighted failed fraction as a function of progenitor mass? Determining this requires either resolving individual progenitors in galaxies beyond the Local Group—which is beyond current capabilities—or measuring the nucleosynthetic yields of successful explosions as a function of progenitor mass with sufficient precision to infer the missing contribution from failed events by comparison with theoretical yield predictions.

Does the Type Ia detonation efficiency evolve with redshift? A systematic comparison of Type Ia host-galaxy metallicity, binary fraction proxies, and light-curve standardization residuals as a function of redshift could in principle constrain $\eta_{\text{Ia}}(z)$ statistically. The Vera Rubin Observatory LSST2009 and the *Euclid* mission Euclid2011 will provide samples large enough to pursue this analysis.

What are the fixed points of the coupled system $(\bar{f}(Z), Z(t))$? The explosion efficiency depends on metallicity, and metallicity is driven by the explosion efficiency. Analyzing the fixed points and stability of this coupled system—which determines whether chemical evolution is self-accelerating, self-limiting, or bistable—would provide a more complete picture of how the present ISM metallicity was reached and how robust it is to perturbations.

Can next-generation gravitational wave observatories constrain the failed-collapse rate at cosmological distances? The Einstein Telescope and Cosmic Explorer Maggiore2020,Evans2021 are projected to detect gravitational wave signals from core collapse in the local universe. Systematic comparison of the gravitational wave event rate with the optically detected supernova rate would constrain f_{fail} independently of electromagnetic surveys, though not the composition-weighted quantity.

The central result in one sentence

The chemical complexity of the universe is the cumulative record of gravitationally bound systems that crossed the dissemination threshold; the systems that did not cross it left no record, their contribution to the interstellar medium is exactly zero, and their base rate is structurally inaccessible to any observational program that relies on detecting their emission—because they have none.

Peer Reviews and Revision History

Round 1 — Reviewer Report

Overall Assessment. The paper is clearly written, intellectually ambitious, and addresses a genuine gap in how astrophysicists think about chemical evolution. The central insight — that synthesis without dissemination is chemically inert — is correct and worth formalizing. However, the paper has significant weaknesses in the gap between its formal apparatus and its physical claims, and in the novelty it attributes to results that are largely implicit in existing literature.

Major Concern 1. The central theorem is formally correct but physically trivial. The claim that a star collapsing to a black hole contributes zero metals to the ISM is not a theorem requiring proof — it is a definition. The paper should either demonstrate that the field has systematically neglected this condition in quantitative models, or use the formalism to derive results that would not be obvious without it.

Major Concern 2. The “epistemological gap” is overstated. Several observational handles exist: the diffuse supernova neutrino background (DSNB) constrains the total core-collapse rate including failed events; the black hole mass function from gravitational wave mergers encodes the failed-SN progenitor mass distribution; galactic chemical evolution inversion already constrains the effective yield-weighted explosion efficiency.

Major Concern 3. The toy models are doing too much work. The quantitative conclusions — factor-of-two fragility, 20 Gyr timescale, 0.2–0.3 dex metallicity shift — derive from hand-tuned toy models with no calibration against 3D simulation results.

Major Concern 4. The relationship to existing literature is underdeveloped. The binary structure of explosion vs. collapse is the foundation of Fryer (1999), O’Connor & Ott (2011), and Sukhbold et al. (2016). Chemical fragility as sensitivity to yield-weighted explosion efficiency is implicit in every closed-box chemical evolution model since Pagel (1997).

Moderate Concern 5. The persistence argument risks circularity: ISM differentiation is defined as the cumulative product of dissemination events, so the existence of a differentiated ISM implies dissemination by construction.

Moderate Concern 6. The Type Ia section conflates two distinct problems: the fraction of near-Chandrasekhar systems that detonate (population completeness) and the redshift evolution of that fraction as a standard-candle systematic (cosmological bias).

Summary recommendation. Major revision required. The author should either substantially strengthen the toy models with calibration against 3D explosion simulations, or retreat

to a purely formal/conceptual paper and remove the quantitative claims that depend on the toy models.

Round 1 — Author Response

We thank the reviewer for a careful and substantive reading. We begin with a statement of purpose that will frame all responses below, because the review consistently assumes that the paper’s goal is to derive new quantitative predictions. It is not.

Why this paper was written. The field of galactic chemical evolution has for decades treated the explosion efficiency as a parameter to be measured, not a phenomenon to be explained. The standard approach — Kobayashi2006, Matteucci1986, and the literature that follows — assumes that some fraction of massive stars explode and proceeds to fit that fraction to abundance data. What is never foregrounded is that this fraction is (a) epistemologically inaccessible in its composition-weighted form, (b) contingent on specific and non-generic features of the particle content of this universe, and (c) the governing variable of chemical complexity across cosmic time. This paper puts that variable at the center. The formal apparatus is the means; the reframing is the purpose.

Response to Major Concern 1 (triviality of the theorem). The reviewer is correct that the conclusion of Theorem 2.6 is implicitly assumed throughout the literature. This is precisely why the theorem needed to be stated. An assumption made without examination cannot be interrogated, generalized, or used as the foundation for a structural argument. The theorem does three things the implicit assumption does not.

First, it makes the binary structure explicit: the contribution is not small, not reduced, not uncertain — it is exactly zero. This exactness is what permits the fragility argument in Section 6.

Second, it places the claim in a framework that applies beyond stellar astrophysics — to any gravitationally bound system in any universe with a closure mechanism. The reviewer treats this as overreach; we treat it as the paper’s main contribution. The theorem is not about supernovae. It is about the structural relationship between closure, dissemination, and the persistence of differentiation.

Third, by formalizing what was implicit, we expose the epistemological gap that the literature has not confronted. If the assumption were explicit in every galactic chemical evolution paper, someone would have asked: what is the uncertainty on the explosion efficiency, can it be measured, and what happens to the metallicity history of the universe if it is a factor of two lower than assumed? No one has asked these questions systematically, because the assumption was never foregrounded.

This is why we write this paper. The reviewer’s objection — that a result obvious in retrospect adds nothing — is the standard objection to all foundational work. Newton’s second law is obvious in retrospect. The point of formalizing it was not to surprise physicists but to provide a foundation from which other results follow. We claim the same status for Theorem 2.6.

Response to Major Concern 2 (epistemological gap overstated). We agree that the DSNB, the black hole mass function, and chemical evolution inversion are genuine observational handles, and we add a paragraph in Section 6.1 crediting them. However, none measures the quantity that governs the rate of ISM chemical enrichment.

The DSNB constrains the total neutrino flux from all core collapses. It therefore constrains the *number* ratio of failed to successful collapses. It does not constrain the composition-weighted failed fraction, because neutrinos carry no nucleosynthetic information. A failed $40 M_{\odot}$ star and a failed $10 M_{\odot}$ star contribute equally to the DSNB but differ by a factor of several in potential iron yield lost behind the horizon.

The black hole mass spectrum from gravitational wave mergers provides information about which progenitors failed — closer to what we need. However, the current sample is biased toward high-mass systems, and the mapping from black hole mass to progenitor composition is not established. More importantly, the merger sample selects black holes that survived in binary systems long enough to merge — a subset with unknown relationship to the full failed-collapse population.

Galactic abundance inversion constrains the effective yield-weighted efficiency, but only as a degenerate combination with the assumed stellar yields. The inversion tells us the efficiency is consistent with assumed yields, not what it is independently.

This is why we write this paper. The composition-weighted failed fraction — the quantity that directly governs chemical evolution — is inaccessible to any current survey. Galactic chemical evolution models carry an unquantified systematic of order unity in their central parameter. Pointing this out is the paper’s epistemological contribution.

Response to Major Concern 3 (toy models doing too much work). The reviewer is correct, and this is the one technical criticism we accept without qualification. In the revised version, all quantitative estimates from toy models are explicitly labelled as order-of-magnitude illustrations. Key numbers are given with ranges reflecting model uncertainty. A calibration note is added comparing toy model results with published 3D simulation results.

We do not, however, remove the quantitative fragility argument. The order-of-magnitude result — a factor-of-two reduction in explosion efficiency approximately doubles the time to reach solar metallicity — is robust to the toy model details. It follows from Proposition 6.2, which is proved from first principles and does not depend on any toy model.

This is why we write this paper. The fragility argument is what connects the formal framework to observational significance. Without it, the paper makes a correct but abstract point. With it, the paper shows that the unobservable parameter controls the chemical history of the universe at the level of cosmic time.

Response to Major Concern 4 (prior art). We add a dedicated “Prior Art and Novelty” paragraph to the Introduction. What is established: explosion efficiency varies with progenitor mass and metallicity Fryer1999,Sukhbold2016; galactic chemical evolution depends on supernova yields Pagel1997,Matteucci1986; failed supernovae exist as candidates Adams2017,Neustadt2021. What is new: (1) the formal identification of the dissemination threshold as the governing variable, distinct from explosion energy or yield; (2) the proof that the composition-weighted contribution of failed events is exactly zero and inaccessible in that specific form; (3) the selection argument: catastrophic phase transitions persist not because they are mandated by physical law but because systems without them leave no differentiated trace; (4) the neutrino section (4): the physical realization of the dissemination threshold in this universe depends on contingent features of the Standard Model — specifically the hierarchy between weak and electromagnetic coupling. A universe without neutrinos would have explosion efficiency exactly zero. This is not in any of the cited literature.

This is why we write this paper. The selection argument at the level of classes of processes — not individual events, not constants of nature — is the conceptual contribution. Explosions are not written into physics. They persist because systems without them are chemically invisible.

Response to Moderate Concern 5 (circularity of persistence argument). The reviewer correctly identifies that the persistence argument is a logical tautology in one sense: if ISM differentiation is defined as the cumulative product of dissemination events, its existence implies dissemination by construction. We accept this. The argument is not presented as an empirical discovery. It is a structural constraint: it specifies what any universe containing a differentiated medium must have contained. The non-trivial content is the *selectional* claim. We observe a differentiated universe not because differentiation is the default outcome, but because undifferentiated universes cannot contain the conditions for making the observation. The argument does not explain why explosion efficiency is as high as it is. It explains why any universe in which we make observations must have positive efficiency. The magnitude is a separate and open question — Section 6 addresses it through the fragility bound.

Response to Moderate Concern 6 (Type Ia conflation). Correct structural observation. Section 5 is revised to explicitly separate: (a) the detonation efficiency as a population completeness problem (what fraction of near-Chandrasekhar systems successfully detonate);

and (b) the redshift evolution of $\eta_{\text{Ia}}(z)$ as a standard-candle systematic (which requires not only that efficiency vary with redshift, but that it covary with standardizable light-curve parameters). A uniform drop in efficiency that does not change the properties of successful events would not bias the distance scale. The cosmological bias argument requires that events dropping out of the sample at high redshift differ systematically in their light-curve properties — which is physically plausible if the DDT probability depends on metallicity, which affects the carbon-to-oxygen ratio and hence the nickel yield.

New material added in revision: Section 4. In response to the general concern that the paper does not explain why this universe contains a dissemination mechanism at all, we add Section 4: “The Neutrino as a Structural Solution to the Closure Problem.” The section argues that explosions are counter-natural — local victories over gravity, thermodynamics, and the inertia of bound systems. The neutrino-driven mechanism works because (a) the weak cross-section is sixteen orders of magnitude smaller than the electromagnetic cross-section, making matter transparent to neutrinos at nuclear density; (b) matter is quantum-structurally mostly empty, so even at nuclear saturation neutrinos are not trapped on macroscopic timescales; (c) the energy scale of collapse produces neutrinos at energies where the mean free path through the gain region matches the gain region depth — the precise scale for effective heating. In a universe without neutrinos, explosion efficiency would be exactly zero. Every core collapse would be a failed supernova. The interstellar medium would retain its primordial composition forever. The neutrino is not a detail of supernova physics. It is the physical realization of the dissemination threshold in this universe — a contingent feature of the Standard Model, not a generic prediction of gravitational physics. This argument is not in any of the cited literature.

References

References

- [Abbott et al.(2017)] Abbott, B. P., et al. (2017). GW170817. *Physical Review Letters*, 119, 161101.
- [Adams et al.(2017)] Adams, S. M., et al. (2017). The search for failed supernovae. *MNRAS*, 469, 1445.
- [Arnett(1996)] Arnett, D. (1996). *Supernovae and Nucleosynthesis*. Princeton University Press.
- [Basinger et al.(2021)] Basinger, C. M., et al. (2021). Failed SN fraction. *MNRAS*, 508, 1156.
- [Beacom(2010)] Beacom, J. F. (2010). The diffuse supernova neutrino background. *Ann. Rev. Nucl. Part. Sci.*, 60, 439.
- [Bethe & Wilson(1985)] Bethe, H. A., & Wilson, J. R. (1985). Revival of a stalled supernova shock by neutrino heating. *ApJ*, 295, 14.
- [Bethe(1990)] Bethe, H. A. (1990). Supernova mechanisms. *Rev. Mod. Phys.*, 62, 801.
- [Burbidge et al.(1957)] Burbidge, E. M., et al. (1957). Synthesis of the elements in stars. *Rev. Mod. Phys.*, 29, 547.
- [Burrows et al.(2021)] Burrows, A., et al. (2021). 3D FORNAX simulations. *MNRAS*, 491, 2715.
- [Casagrande et al.(2011)] Casagrande, L., et al. (2011). Chemical evolution of the solar neighbourhood. *A&A*, 530, A138.
- [Cowan et al.(2021)] Cowan, J. J., et al. (2021). The rapid neutron-capture process. *Rev. Mod. Phys.*, 93, 015002.
- [Cromartie et al.(2020)] Cromartie, H. T., et al. (2020). Shapiro delay measurements. *Nature Astronomy*, 4, 72.
- [Curti et al.(2020)] Curti, M., et al. (2020). The mass-metallicity relation revisited. *MNRAS*, 491, 944.
- [Demorest et al.(2010)] Demorest, P. B., et al. (2010). A two-solar-mass neutron star. *Nature*, 467, 1081.

- [Dessart et al.(2006)] Dessart, L., et al. (2006). Accretion-induced collapse of white dwarfs. *ApJ*, 644, 1063.
- [Euclid Collaboration(2011)] Euclid Collaboration (2011). Euclid Definition Study Report. *arXiv:1110.3193*.
- [Evans et al.(2021)] Evans, M., et al. (2021). A Horizon Study for Cosmic Explorer. *arXiv:2109.09882*.
- [Fink et al.(2014)] Fink, M., et al. (2014). Thermonuclear supernovae. *MNRAS*, 438, 1762.
- [Fryer(1999)] Fryer, C. L. (1999). Mass limits for black hole formation. *ApJ*, 522, 413.
- [Gerke et al.(2015)] Gerke, J. R., et al. (2015). First failed SN candidates. *MNRAS*, 450, 3289.
- [Gossan et al.(2016)] Gossan, S. E., et al. (2016). GW from core-collapse SNe. *Phys. Rev. D*, 93, 042002.
- [Graur et al.(2013)] Graur, O., et al. (2013). Supernovae in the Subaru Deep Field. *ApJ*, 783, 28.
- [Guy et al.(2007)] Guy, J., et al. (2007). SALT2. *A&A*, 466, 11.
- [Heger et al.(2003)] Heger, A., et al. (2003). How massive single stars end their life. *ApJ*, 591, 288.
- [Hillebrandt et al.(2013)] Hillebrandt, W., et al. (2013). Type Ia supernovae. *Frontiers of Physics*, 8, 116.
- [Iwamoto et al.(1999)] Iwamoto, K., et al. (1999). Nucleosynthesis in Chandrasekhar mass models. *ApJS*, 125, 439.
- [Janka(2001)] Janka, H.-T. (2001). Conditions for shock revival. *A&A*, 368, 527.
- [Janka(2012)] Janka, H.-T. (2012). Explosion mechanisms of core-collapse supernovae. *Ann. Rev. Nucl. Part. Sci.*, 62, 407.
- [Janka(2016)] Janka, H.-T. (2016). Neutrino-driven explosions. *Handbook of Supernovae*. Springer.
- [Kobayashi et al.(2006)] Kobayashi, C., et al. (2006). Chemical diversity of dwarf elliptical galaxies. *ApJ*, 653, 1145.

- [Kochanek et al.(2008)] Kochanek, C. S., et al. (2008). Monitoring a million supergiants for failed supernovae. *ApJ*, 684, 1336.
- [Kriger(2024)] Kriger, B. (2024). The transformational basis of persistence. <https://doi.org/10.5281/zenodo.18435982>
- [Kriger(2026)] Kriger, B. (2026). Structural genesis of dynamical architecture. <https://doi.org/10.5281/zenodo.18646861>
- [Krueger et al.(2010)] Krueger, B. K., et al. (2010). Metallicity and Type Ia supernovae. *ApJ*, 719, 1067.
- [Lodders(2003)] Lodders, K. (2003). Solar system abundances. *ApJ*, 591, 1220.
- [Lovegrove & Woosley(2013)] Lovegrove, E., & Woosley, S. E. (2013). Very low energy supernovae. *ApJ*, 769, 109.
- [LSST Science Collaboration(2009)] LSST Science Collaboration (2009). LSST Science Book. *arXiv:0912.0201*.
- [Maggiore et al.(2020)] Maggiore, M., et al. (2020). Science case for the Einstein Telescope. *JCAP*, 2020, 050.
- [Maiolino & Mannucci(2019)] Maiolino, R., & Mannucci, F. (2019). De re metallica. *A&ARv*, 27, 3.
- [Mannucci et al.(2010)] Mannucci, F., et al. (2010). Mass, star formation rate and metallicity. *MNRAS*, 408, 2115.
- [Maoz & Mannucci(2012)] Maoz, D., & Mannucci, F. (2012). Type-Ia supernova rates. *PASA*, 29, 447.
- [Maoz et al.(2014)] Maoz, D., et al. (2014). Progenitors of Type Ia supernovae. *ARA&A*, 52, 107.
- [Matteucci & Greggio(1986)] Matteucci, F., & Greggio, L. (1986). Type I and II supernovae in chemical enrichment. *A&A*, 154, 279.
- [Melson et al.(2015)] Melson, T., et al. (2015). Neutrino-driven supernova in 3D. *ApJL*, 801, L24.
- [Moe & Di Stefano(2017)] Moe, M., & Di Stefano, R. (2017). Binary star distributions. *ApJS*, 230, 15.

- [Müller(2016)] Müller, B. (2016). Dynamics of neutrino-driven explosions. *MNRAS*, 453, 287.
- [Müller(2019)] Müller, B. (2019). Core-collapse supernovae multi-messenger signals. *Ann. Rev. Nucl. Part. Sci.*, 69, 253.
- [Neustadt et al.(2021)] Neustadt, J. M. M., et al. (2021). Disappearing star candidate. *MNRAS*, 508, 516.
- [Nomoto et al.(1984)] Nomoto, K., et al. (1984). Accreting white dwarf models. *ApJ*, 286, 644.
- [Nomoto & Kondo(1991)] Nomoto, K., & Kondo, Y. (1991). Accretion-induced collapse of white dwarfs. *ApJL*, 367, L19.
- [O’Connor & Ott(2011)] O’Connor, E., & Ott, C. D. (2011). Black hole formation in failing CCSNe. *ApJ*, 730, 70.
- [Ott(2009)] Ott, C. D. (2009). GW signature of core-collapse supernovae. *CQG*, 26, 063001.
- [Pagel(1997)] Pagel, B. E. J. (1997). *Nucleosynthesis and Chemical Evolution of Galaxies*. Cambridge University Press.
- [Perlmutter et al.(1999)] Perlmutter, S., et al. (1999). Measurements of Ω and Λ . *ApJ*, 517, 565.
- [Phillips(1993)] Phillips, M. M. (1993). Absolute magnitudes of Type Ia supernovae. *ApJL*, 413, L105.
- [Prantzos et al.(2018)] Prantzos, N., et al. (2018). Chemical evolution with rotating massive star yields. *MNRAS*, 476, 3432.
- [Reynolds et al.(2015)] Reynolds, T. M., et al. (2015). Progenitors of supernovae and failed supernovae. *ApJ*, 808, 5.
- [Riess et al.(1998)] Riess, A. G., et al. (1998). Observational evidence for an accelerating universe. *AJ*, 116, 1009.
- [Roepke et al.(2007)] Röpke, F. K., et al. (2007). Deflagration-to-detonation transition. *ApJL*, 668, L99.
- [Ruiter et al.(2009)] Ruiter, A. J., et al. (2009). Rates and delay times of Type Ia supernovae. *ApJ*, 699, 2026.

- [Ruiter(2019)] Ruiter, A. J. (2019). Type Ia supernova progenitor origin. *IAU Symposium*, 357, 1.
- [Seitenzahl et al.(2013)] Seitenzahl, I. R., et al. (2013). Delayed-detonation models. *MNRAS*, 429, 1156.
- [Shapiro & Teukolsky(1983)] Shapiro, S. L., & Teukolsky, S. A. (1983). *Black Holes, White Dwarfs, and Neutron Stars*. Wiley.
- [Smartt(2009)] Smartt, S. J. (2009). Progenitors of core-collapse supernovae. *ARA&A*, 47, 63.
- [Sukhbold et al.(2016)] Sukhbold, T., et al. (2016). Core-collapse supernovae from 9 to 120 solar masses. *ApJ*, 821, 38.
- [Timmes et al.(2003)] Timmes, F. X., et al. (2003). The $^{12}\text{C}+^{12}\text{C}$ rate and nucleosynthesis. *ApJ*, 584, 1048.
- [Toonen et al.(2012)] Toonen, S., et al. (2012). SeBa population synthesis code. *A&A*, 546, A70.
- [Ugliano et al.(2012)] Ugliano, M., et al. (2012). Progenitor-explosion connection. *ApJ*, 757, 69.
- [VanDyk(2017)] Van Dyk, S. D. (2017). Progenitors of core-collapse supernovae. *Phil. Trans. R. Soc. A*, 375, 20160277.
- [Vink et al.(2001)] Vink, J. S., et al. (2001). Mass-loss rates of very massive stars. *A&A*, 369, 574.
- [Woosley & Janka(2005)] Woosley, S., & Janka, T. (2005). Physics of core-collapse supernovae. *Nature Physics*, 1, 147.